MAHATMA EDUCATION SOCIETY'S

PILLAI COLLEGE OF ARTS, COMMERCE & SCIENCE
(Empowered Autonomous)

NEW PANVEL

PROJECT REPORT ON

"QUEUEFLOW ANALYTICS – WAIT-TIME OPTIMIZATION"

IN PARTIAL FULFILLMENT OF

M.Sc. DATA ANALYTICS

SEMESTER I– 2026-27

PROJECT GUIDE

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Mahatma Education Society's

PILLAI COLLEGE OF ARTS,COMMERCE & SCIENCE

(Empowered Autonomous)
Re-accredited "A" Grade by NAAC (3rd Cycle)

Project Completion Certificate

THIS IS TO CERTIFY THAT

MOIN SHAIKH

of **M.Sc. Data Analytics Part - I** has completed the project titled **QueueFlow Analytics – Wait-Time Optimization** of subject **Data Analytics with Advanced Excel** under our guidance and supervision during the academic year 2026-27 in the department of Computer Science.

Project Guide	Course Coordinator	Head of the Department

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ABSTRACT

QueueFlow Analytics – Wait-Time Optimization is an Excel-based data analytics project developed to analyze and optimize queue waiting times in a service environment. The project uses a structured dataset containing queue transactions, customer details, service information, counter details, waiting time, service time, payment information and customer satisfaction data.

The project follows a complete data analytics workflow, beginning with raw and inconsistent data inspection and cleaning. Advanced Excel formulas and analytical techniques are used to transform the raw data into meaningful information. The project further applies forecasting, trend analysis, Power Query, What-If Analysis, Power Pivot and DAX to identify patterns and performance indicators related to queue management.

The final stage presents the analyzed information through an interactive Excel dashboard containing key performance indicators, charts, filters and visualizations. The project demonstrates how Advanced Excel can be used to convert raw operational data into useful insights for understanding waiting-time patterns and supporting data-driven queue management decisions.

INTRODUCTION

In many service environments, customers have to wait in queues before receiving the required service. Long waiting times can affect customer satisfaction, service efficiency and the overall performance of an organization. Therefore, analyzing queue-related data can help identify waiting-time patterns, service bottlenecks and areas where operational improvements may be required.

QueueFlow Analytics – Wait-Time Optimization is developed as a data analytics project using Advanced Excel to study queue management and waiting-time patterns. The project works with raw queue transaction data containing information such as queue IDs, dates, times, locations, services, counters, waiting time, service time, customer type, payment mode and satisfaction ratings.

The project follows a systematic data analytics process. Initially, the raw dataset is inspected for missing values, duplicate records, inconsistent formats and invalid entries. The data is then cleaned and transformed using Excel functions and analytical techniques. Further analysis is performed using forecasting, Power Query, What-If Analysis, Power Pivot and DAX.

Finally, the processed data is presented through an interactive dashboard containing key performance indicators and visualizations. The project demonstrates the practical application of Advanced Excel concepts to transform raw operational data into meaningful insights for queue and service management.

OBJECTIVES

1. **To inspect and understand** the raw queue dataset and identify data quality issues such as missing values, duplicate records, inconsistent formats and invalid entries.

2. **To clean and transform** the raw queue data using Advanced Excel functions and techniques.

3. **To analyze queue waiting time and service time** across different services, locations, counters and customer categories.

4. **To apply advanced Excel formulas** for lookup, logical analysis, dynamic data handling and error management.

5. **To perform forecasting and trend analysis** to understand patterns in queue waiting times.

6. **To use Power Query** for data transformation, filtering, grouping, merging and other ETL operations.

7. **To apply What-If Analysis** using Scenario Manager, Goal Seek and Data Tables for queue management situations.

8. **To develop a Power Pivot data model and DAX measures** for advanced performance analysis.

9. **To create an interactive dashboard** using PivotTables, PivotCharts, KPIs, slicers and other visualizations.

10. **To demonstrate VBA automation** for simplifying repetitive reporting activities.

DATASET DESCRIPTION

The QueueFlow Analytics project uses a raw dataset containing **10,000 queue transaction records**. Each record represents a customer queue transaction and contains information related to the queue, date and time, service, counter, location, waiting time, service time, customer category, payment details and satisfaction.

The dataset has been intentionally prepared with different data-quality issues to simulate a real-world operational dataset. These issues include missing values, duplicate Queue IDs, inconsistent data formats and invalid entries. The raw dataset is used as the starting point for the data cleaning and transformation process.

The dataset contains **20 columns**, which provide the required information for performing queue analysis, service performance analysis, forecasting and dashboard development.

Dataset Fields

Sr. No.	Field	Description
1	Queue_ID	Unique identifier of a queue transaction
2	Date	Date of the queue transaction
3	Time	Time at which the queue transaction occurred
4	Location_ID	Identifier of the service location
5	Location	Name of the service location
6	Service_ID	Identifier of the service
7	Service	Type of service requested
8	Counter_ID	Identifier of the service counter
9	Counter	Service counter information
10	Customer_Type	Category of customer
11	Waiting_Time	Time spent waiting in the queue
12	Service_Time	Time taken to provide the service
13	Status	Status of the queue transaction
14	Payment_Mode	Mode of payment used
15	Service_Fee	Fee associated with the service
16	Satisfaction	Customer satisfaction rating
17	Staff_ID	Identifier of the staff member
18	Priority	Priority level of the customer
19	Day	Day associated with the transaction
20	Remarks	Additional information related to the transaction

TOOLS & TECHNOLOGIES USED

The QueueFlow Analytics project is developed using Microsoft Excel and its advanced data analytics features. Different Excel tools and techniques are used at different stages of the project, starting from raw data cleaning and formula-based analysis to advanced data modelling, visualization and automation.

5.1 Microsoft Excel

Microsoft Excel is the primary tool used for data cleaning, analysis, transformation, visualization and dashboard development. Advanced Excel functions are used to process the queue dataset and generate meaningful analytical results.

5.2 Excel Functions

Lookup, reference, logical, dynamic array, statistical, financial and date/time functions are used to perform data transformation and analytical calculations.

5.3 Power Query

Power Query is used for importing, cleaning and transforming data. Operations such as filtering, sorting, removing duplicates, merging, grouping and reshaping data are applied where required.

5.4 Power Pivot & DAX

Power Pivot is used to create relationships between different datasets and build a data model. DAX measures are used to calculate key performance indicators and analytical metrics.

5.5 PivotTables & PivotCharts

PivotTables and PivotCharts are used to summarize queue data and visually analyze waiting time, service performance and other operational metrics.

5.6 What-If Analysis

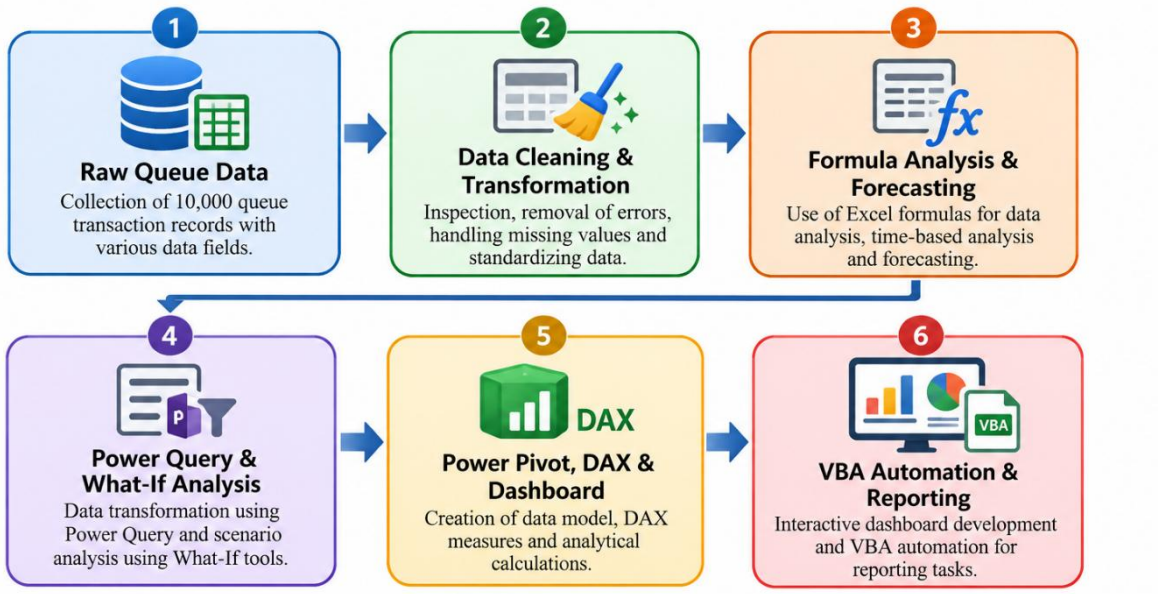
What-If Analysis techniques such as Scenario Manager, Goal Seek and Data Tables are used to evaluate different queue-management situations.

5.7 Dashboard & VBA

An interactive dashboard is developed using charts, KPIs, slicers and other Excel visualization features. VBA automation is used to simplify repetitive reporting tasks.

PROJECT METHODOLOGY

The QueueFlow Analytics project follows a structured data analytics methodology. The process begins with the inspection of raw queue transaction data and continues through data cleaning, formula-based analysis, advanced data transformation, forecasting, data modelling and dashboard development. Each stage is designed to progressively convert raw operational data into meaningful analytical information.

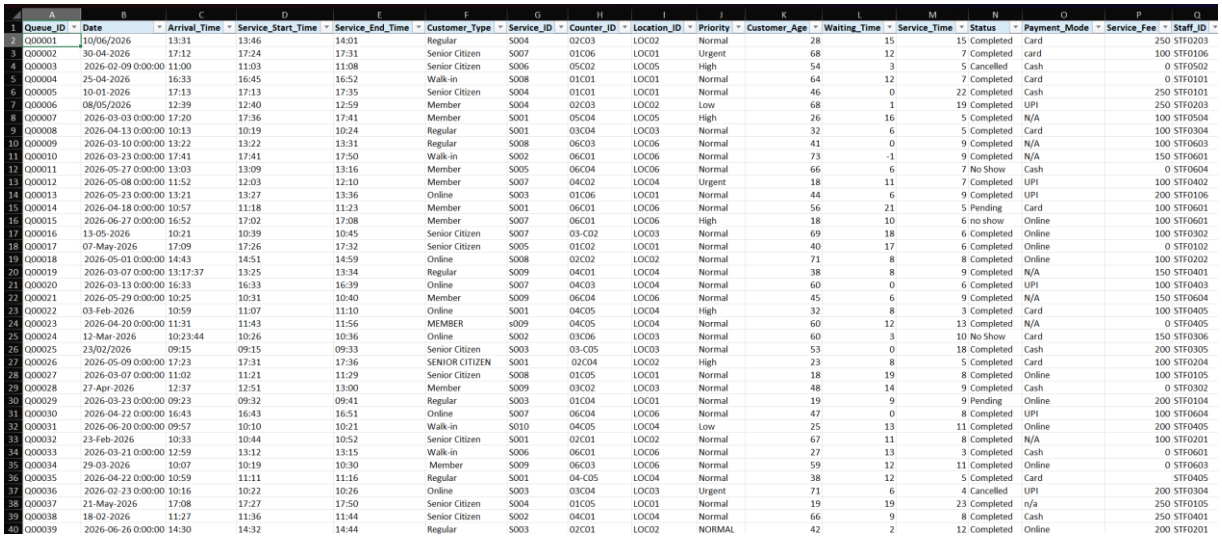


DATA CLEANING & FORMULA ANALYSIS

Raw Data Inspection

The raw QueueFlow dataset was inspected before performing any data cleaning or transformation. The dataset contains queue transaction records with fields related to queue identification, date and time, service, customer type, service and counter information, priority, waiting time, service time, status, payment mode, service fee, staff information, customer satisfaction, complaints and remarks.

During the initial inspection, several data-quality issues were observed in the raw dataset. These include inconsistent date formats, inconsistent capitalization in categorical values, different formats of IDs, blank or missing values, and entries that require validation before analysis. These issues need to be addressed before performing further calculations and analytical operations.



Queue_ID	Date	Arrival_Time	Service_Start_Time	Service_End_Time	Customer_Type	Service_ID	Counter_ID	Location_ID	Priority	Customer_Age	Waiting_Time	Service_Time	Status	Payment_Mode	Service_Fee	Staff_ID
Q00001	10/06/2026	13:31	13:46	14:01	Regular	S004	02C03	LOC02	Normal	28	15	15	Completed	Card	250	STF0203
Q00002	30-04-2026	17:12	17:24	17:31	Senior Citizen	S007	01C06	LOC01	Urgent	68	12	7	Completed	card	100	STF0106
Q00003	2026-02-09 0:00:00	11:00	11:03	11:08	Senior Citizen	S006	05C02	LOC05	High	54	3	5	Cancelled	Cash	0	STF0502
Q00004	25-04-2026	16:33	16:45	16:52	Walk-in	S008	01C01	LOC01	Normal	64	12	7	Completed	Card	0	STF0101
Q00005	10-01-2026	17:13	17:13	17:35	Senior Citizen	S004	01C01	LOC01	Normal	46	0	22	Completed	Cash	250	STF0101
Q00006	08/05/2026	12:39	12:40	12:59	Member	S004	02C03	LOC02	Low	68	1	19	Completed	UPI	250	STF0203
Q00007	2026-03-03 0:00:00	17:20	17:36	17:41	Member	S001	05C04	LOC05	High	26	16	5	Completed	N/A	100	STF0504
Q00008	2026-04-13 0:00:00	10:13	10:19	10:24	Regular	S001	03C04	LOC03	Normal	32	6	5	Completed	Card	100	STF0304
Q00009	2026-03-10 0:00:00	13:22	13:22	13:31	Regular	S008	06C03	LOC06	Normal	41	0	9	Completed	N/A	100	STF0603
Q00010	2026-03-23 0:00:00	17:41	17:41	17:50	Walk-in	S002	06C01	LOC06	Normal	73	-1	9	Completed	N/A	150	STF0601
Q00011	2026-05-27 0:00:00	13:03	13:09	13:16	Member	S005	06C04	LOC06	Normal	66	6	7	No Show	Cash	0	STF0604
Q00012	2026-05-08 0:00:00	11:52	12:03	12:10	Member	S007	04C02	LOC04	Urgent	18	11	7	Completed	UPI	100	STF0402
Q00013	2026-05-23 0:00:00	13:21	13:27	13:36	Online	S003	01C06	LOC01	Normal	44	6	9	Completed	UPI	200	STF0106
Q00014	2026-04-18 0:00:00	10:57	11:18	11:23	Member	S001	06C01	LOC06	Normal	56	21	5	Pending	Card	100	STF0601
Q00015	2026-06-27 0:00:00	16:52	17:02	17:08	Member	S007	06C01	LOC06	High	18	10	6	no show	Online	100	STF0601
Q00016	13-05-2026	10:21	10:39	10:45	Senior Citizen	S007	03-C02	LOC03	Normal	69	18	6	Completed	Online	100	STF0302
Q00017	07-May-2026	17:09	17:26	17:32	Senior Citizen	S005	01C02	LOC01	Normal	40	17	6	Completed	Online	0	STF0102
Q00018	2026-05-01 0:00:00	14:43	14:51	14:59	Online	S008	02C02	LOC02	Normal	71	8	8	Completed	Online	100	STF0202
Q00019	2026-03-07 0:00:00	13:17:37	13:25	13:34	Regular	S009	04C01	LOC04	Normal	38	8	9	Completed	N/A	150	STF0401
Q00020	2026-03-13 0:00:00	16:33	16:33	16:39	Online	S007	04C03	LOC04	Normal	60	0	6	Completed	UPI	100	STF0403
Q00021	2026-05-29 0:00:00	10:25	10:31	10:40	Member	S009	06C04	LOC06	Normal	45	6	9	Completed	N/A	150	STF0604
Q00022	03-Feb-2026	10:59	11:07	11:10	Online	S001	04C05	LOC04	High	32	8	3	Completed	Card	100	STF0405
Q00023	2026-04-20 0:00:00	11:31	11:43	11:56	MEMBER	S009	04C05	LOC04	Normal	60	12	13	Completed	N/A	0	STF0405
Q00024	12-Mar-2026	10:23:44	10:26	10:36	Online	S002	03C06	LOC03	Normal	60	3	10	No Show	Card	150	STF0306
Q00025	23/02/2026	09:15	09:15	09:33	Senior Citizen	S003	03-C05	LOC03	Normal	53	0	18	Completed	Cash	200	STF0305
Q00026	2026-05-09 0:00:00	17:23	17:31	17:36	SENIOR CITIZEN	S001	02C04	LOC02	High	23	8	5	Completed	Card	100	STF0204
Q00027	2026-03-07 0:00:00	11:02	11:21	11:29	Senior Citizen	S008	01C05	LOC01	Normal	18	19	8	Completed	Online	100	STF0105
Q00028	27-Apr-2026	12:37	12:51	13:00	Member	S009	03C02	LOC03	Normal	48	14	9	Completed	Cash	0	STF0302
Q00029	2026-03-23 0:00:00	09:23	09:32	09:41	Regular	S003	01C04	LOC01	Normal	19	9	9	Pending	Online	200	STF0104
Q00030	2026-04-22 0:00:00	16:43	16:43	16:51	Online	S007	06C04	LOC06	Normal	47	0	8	Completed	UPI	100	STF0604
Q00031	2026-06-20 0:00:00	09:57	10:10	10:21	Walk-in	S010	04C05	LOC04	Low	25	13	11	Completed	Online	200	STF0405
Q00032	23-Feb-2026	10:33	10:44	10:52	Senior Citizen	S001	02C01	LOC02	Normal	67	11	8	Completed	N/A	100	STF0201
Q00033	2026-03-21 0:00:00	12:59	13:12	13:15	Walk-in	S006	06C01	LOC06	Normal	27	13	3	Completed	Cash	0	STF0601
Q00034	29-03-2026	10:07	10:19	10:30	Member	S009	06C03	LOC06	Normal	59	12	11	Completed	Online	0	STF0603
Q00035	2026-04-22 0:00:00	10:59	11:11	11:16	Regular	S001	04-C05	LOC04	Normal	38	12	5	Completed	Card		STF0405
Q00036	2026-02-23 0:00:00	10:16	10:22	10:26	Online	S003	03C04	LOC03	Urgent	71	6	4	Cancelled	UPI	200	STF0304
Q00037	21-May-2026	17:08	17:27	17:50	Senior Citizen	S004	01C05	LOC01	Normal	19	19	23	Completed	n/a	250	STF0105
Q00038	18-02-2026	11:27	11:36	11:44	Senior Citizen	S002	04C01	LOC04	Normal	66	9	8	Completed	Cash	250	STF0401
Q00039	2026-06-26 0:00:00	14:30	14:32	14:44	Regular	S003	02C01	LOC02	NORMAL	42	2	12	Completed	Online	200	STF0201

Observed Data Quality Issues

Issue	Example Observed
Inconsistent Date Format	10/06/2026, 30-04-2026, 07-May-2026
Display Issue in Date Column	########## visible in several date cells
Inconsistent Capitalization	Member, MEMBER, Senior Citizen, SENIOR CITIZEN
Inconsistent Payment Mode	Card, card, Cash, cash, UPI, N/A
Inconsistent Status	Completed, completed, No Show, no show, Pending
Missing/Blank Values	Some fields may contain blank values
Data Validation Required	Waiting time, service time and other numerical fields require validation

MODULE 1 – DATA CLEANING & FORMULA ANALYSIS

7.1 Module Overview

Module 1 of the QueueFlow Analytics project focuses on preparing and analysing queue data using advanced Excel formulas and data-cleaning techniques. The raw queue dataset contained 10,000 records with inconsistent formats, missing values, duplicate Queue IDs and invalid values.

The data was transformed into a structured `Cleaned_Data` sheet. Excel functions such as `IF`, `OR`, `AND`, `IFERROR`, `COUNTIF`, `AVERAGEIF`, `SUMPRODUCT`, `UNIQUE`, `FILTER`, `SORT`, `SORTBY`, `INDEX`, `MATCH` and `XLOOKUP` were used for cleaning, classification, lookup and analysis.

7.2 Data Cleaning

Raw data contained inconsistent formats, missing values, duplicate IDs and invalid values. A separate `Cleaned_Data` sheet was created to standardize and validate the data.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Queue_ID	Date	Arrival_Time	Service_Start_Time	Service_End_Time	Customer_Type	Service_ID	Counter_ID	Location_ID	Priority	Customer_Age	Waiting_Time	Service_Time	Status	Payment_Mode
2	Q00001	10/06/2026	13:31	13:46	14:01	Regular	S004	02C03	LOC02	Normal	28	15	15	Completed	Card
3	Q00002	30-04-2026	17:12	17:24	17:31	Senior Citizen	S007	01C06	LOC01	Urgent	68	12	7	Completed	card
4	Q00003	2026-02-09 0:00:00	11:00	11:03	11:08	Senior Citizen	S006	05C02	LOC05	High	54	3	5	Cancelled	Cash
5	Q00004	25-04-2026	16:33	16:45	16:52	Walk-in	S008	01C01	LOC01	Normal	64	12	7	Completed	Card
6	Q00005	10-01-2026	17:13	17:13	17:35	Senior Citizen	S004	01C01	LOC01	Normal	46	0	22	Completed	Cash
7	Q00006	08/05/2026	12:39	12:40	12:59	Member	S004	02C03	LOC02	Low	68	1	19	Completed	UPI
8	Q00007	2026-03-03 0:00:00	17:20	17:36	17:41	Member	S001	05C04	LOC05	High	26	16	5	Completed	N/A
9	Q00008	2026-04-13 0:00:00	10:13	10:19	10:24	Regular	S001	03C04	LOC03	Normal	32	6	5	Completed	Card
10	Q00009	2026-03-10 0:00:00	13:22	13:22	13:31	Regular	S008	06C03	LOC06	Normal	41	0	9	Completed	N/A
11	Q00010	2026-03-23 0:00:00	17:41	17:41	17:50	Walk-in	S002	06C01	LOC06	Normal	73	-1	9	Completed	N/A
12	Q00011	2026-05-27 0:00:00	13:03	13:09	13:16	Member	S005	06C04	LOC06	Normal	66	6	7	No Show	Cash
13	Q00012	2026-05-08 0:00:00	11:52	12:03	12:10	Member	S007	04C02	LOC04	Urgent	18	11	7	Completed	UPI
14	Q00013	2026-05-23 0:00:00	13:21	13:27	13:36	Online	S003	01C06	LOC01	Normal	44	6	9	Completed	UPI
15	Q00014	2026-04-18 0:00:00	10:57	11:18	11:23	Member	S001	06C01	LOC06	Normal	56	21	5	Pending	Card
16	Q00015	2026-06-27 0:00:00	16:52	17:02	17:08	Member	S007	06C01	LOC06	High	18	10	6	no show	Online
17	Q00016	13-05-2026	10:21	10:39	10:45	Senior Citizen	S007	03-C02	LOC03	Normal	69	18	6	Completed	Online
18	Q00017	07-May-2026	17:09	17:26	17:32	Senior Citizen	S005	01C02	LOC01	Normal	40	17	6	Completed	Online
19	Q00018	2026-05-01 0:00:00	14:43	14:51	14:59	Online	S008	02C02	LOC02	Normal	71	8	8	Completed	Online
20	Q00019	2026-03-07 0:00:00	13:17:37	13:25	13:34	Regular	S009	04C01	LOC04	Normal	38	8	9	Completed	N/A
21	Q00020	2026-03-13 0:00:00	16:33	16:33	16:39	Online	S007	04C03	LOC04	Normal	60	0	6	Completed	UPI
22	Q00021	2026-05-29 0:00:00	10:25	10:31	10:40	Member	S009	06C04	LOC06	Normal	45	6	9	Completed	N/A
23	Q00022	03-Feb-2026	10:59	11:07	11:10	Online	S001	04C05	LOC04	High	32	8	3	Completed	Card
24	Q00023	2026-04-20 0:00:00	11:31	11:43	11:56	MEMBER	s009	04C05	LOC04	Normal	60	12	13	Completed	N/A
25	Q00024	12-Mar-2026	10:23:44	10:26	10:36	Online	S002	03C06	LOC03	Normal	60	3	10	No Show	Card
26	Q00025	23/02/2026	09:15	09:15	09:33	Senior Citizen	S003	03-C05	LOC03	Normal	53	0	18	Completed	Cash
27	Q00026	2026-05-09 0:00:00	17:23	17:31	17:36	SENIOR CITIZEN	S001	02C04	LOC02	High	23	8	5	Completed	Card
28	Q00027	2026-03-07 0:00:00	11:02	11:21	11:29	Senior Citizen	S008	01C05	LOC01	Normal	18	19	8	Completed	Online
29	Q00028	27-Apr-2026	12:37	12:51	13:00	Member	S009	03C02	LOC03	Normal	48	14	9	Completed	Cash
30	Q00029	2026-03-23 0:00:00	09:23	09:32	09:41	Regular	S003	01C04	LOC01	Normal	19	9	9	Pending	Online
31	Q00030	2026-04-22 0:00:00	16:43	16:43	16:51	Online	S007	06C04	LOC06	Normal	47	0	8	Completed	UPI
32	Q00031	2026-06-20 0:00:00	09:57	10:10	10:21	Walk-in	S010	04C05	LOC04	Low	25	13	11	Completed	Online
33	Q00032	23-Feb-2026	10:33	10:44	10:52	Senior Citizen	S001	02C01	LOC02	Normal	67	11	8	Completed	N/A
34	Q00033	2026-03-21 0:00:00	12:59	13:12	13:15	Walk-in	S006	06C01	LOC06	Normal	27	13	3	Completed	Cash
35	Q00034	29-03-2026	10:07	10:19	10:30	Member	S009	06C03	LOC06	Normal	59	12	11	Completed	Online
36	Q00035	2026-04-22 0:00:00	10:59	11:11	11:16	Regular	S001	04-C05	LOC04	Normal	38	12	5	Completed	Card
37	Q00036	2026-02-23 0:00:00	10:16	10:22	10:26	Online	S003	03C04	LOC03	Urgent	71	6	4	Cancelled	UPI
38	Q00037	21-May-2026	17:08	17:27	17:50	Senior Citizen	S004	01C05	LOC01	Normal	19	19	23	Completed	n/a
39	Q00038	18-02-2026	11:27	11:36	11:44	Senior Citizen	S002	04C01	LOC04	Normal	66	9	8	Completed	Cash
40	Q00039	2026-06-26 0:00:00	14:30	14:32	14:44	Regular	S003	02C01	LOC02	NORMAL	42	2	12	Completed	Online
41	Q00040	2026-05-02 0:00:00	12:03	12:19	12:32	Regular	S005	01C04	loc01	Normal	35	16	13	Completed	Cash
42	Q00041	07-02-2026	10:45	10:56	11:03	Online	S007	03C04	LOC03	Normal	44	11	7	Completed	card
43	Q00042	2026-02-13 0:00:00	16:41		17:06	Member	S003	01C06	LOC01	Normal	62	16	9	Completed	Online
44	Q00043	14-May-2026	12:03	12:08	12:16	Member	S002	05C05	LOC05	High	57	5	8	Completed	Cash

Project_Setup | Raw_Queue_Data | Cleaned_Data | Data_Validation | Module1_Analysis | Service_Master | Location_Mast

Figure 7.1: *Raw Queue Data – Columns A to O*

P	Q	R	S	T
Service_Fee	Staff_ID	Satisfaction_Score	Complaint	Remarks
250	STF0203	5	None	Priority case
100	STF0106	4	None	Needs review
0	STF0502	5	None	
0	STF0101	5	None	Priority case
250	STF0101	5	None	Priority case
250	STF0203	3	None	Verified
100	STF0504	5	None	Verified
100	STF0304	4	Payment Issue	Verified
100	STF0603	4	None	Needs review
150	STF0601	4	None	Needs review
0	STF0604	5	None	Priority case
100	STF0402	1	None	
200	STF0106	3	None	Priority case
100	STF0601	1	Service Issue	Verified
100	STF0601	2	None	Follow-up
100	STF0302	2	Payment Issue	Follow-up
0	STF0102	2	None	Verified
100	STF0202	5	None	
150	STF0401	1	None	Needs review
100	STF0403	1	None	
150	STF0604	1	None	Verified
100	STF0405	5	None	Needs review
0	STF0405	3	None	Verified
150	STF0306	2	None	Needs review
200	STF0305	4	None	Verified
100	STF0204	1	None	Priority case
100	STF0105	4	None	Needs review
0	STF0302	3	None	Follow-up
200	STF0104	1	Payment Issue	Follow-up
100	STF0604	3	None	Needs review
200	STF0405	3	None	Verified
100	STF0201	3	None	Needs review
0	STF0601	2	None	Follow-up
0	STF0603	1	Service Issue	
	STF0405	2	Staff Issue	
200	STF0304	4	None	Verified
250	STF0105	1	Service Issue	Follow-up
250	STF0401	5	Long Wait	Verified
200	STF0201	5	None	Follow-up
0	STF0104	1	None	
100	STF0304	5	None	Needs review
200	STF0106	5	Long Wait	Priority case
150	STF0505	1	None	Follow-up

Figure 7.2: *Raw Queue Data – Columns P to T*

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Queue_ID	Date	Arrival_Time	Service_Start_Time	Service_End_Time	Customer_Type	Service_ID	Counter_ID	Location_ID	Priority	Customer_Age	Waiting_Time	Service_Time	Status	Payment_Mode
Q00001	10-06-2026	13:31:00	13:46:00	14:01:00	Regular	S004	02C03	LOC02	Normal	28	15	15	Completed	Card
Q00002	30-04-2026	17:12:00	17:24:00	17:31:00	Senior Citizen	S007	01C06	LOC01	Urgent	68	12	7	Completed	Card
Q00003	09-02-2026	11:00:00	11:03:00	11:08:00	Senior Citizen	S006	05C02	LOC05	High	54	3	5	Cancelled	Cash
Q00004	25-04-2026	16:33:00	16:45:00	16:52:00	Walk-in	S008	01C01	LOC01	Normal	64	12	7	Completed	Card
Q00005	10-01-2026	17:13:00	17:13:00	17:35:00	Senior Citizen	S004	01C01	LOC01	Normal	46	0	22	Completed	Cash
Q00006	08-05-2026	12:39:00	12:40:00	12:59:00	Member	S004	02C03	LOC02	Low	68	1	19	Completed	UPI
Q00007	03-03-2026	17:20:00	17:36:00	17:41:00	Member	S001	05C04	LOC05	High	26	16	5	Completed	N/A
Q00008	13-04-2026	10:13:00	10:19:00	10:24:00	Regular	S001	03C04	LOC03	Normal	32	6	5	Completed	Card
Q00009	10-03-2026	13:22:00	13:22:00	13:31:00	Regular	S008	06C03	LOC06	Normal	41	0	9	Completed	N/A
Q00010	23-03-2026	17:41:00	17:41:00	17:50:00	Walk-in	S002	06C01	LOC06	Normal	73		9	Completed	N/A
Q00011	27-05-2026	13:03:00	13:09:00	13:16:00	Member	S005	06C04	LOC06	Normal	66	6	7	No Show	Cash
Q00012	08-05-2026	11:52:00	12:03:00	12:10:00	Member	S007	04C02	LOC04	Urgent	18	11	7	Completed	UPI
Q00013	23-05-2026	13:21:00	13:27:00	13:36:00	Online	S003	01C06	LOC01	Normal	44	6	9	Completed	UPI
Q00014	18-04-2026	10:57:00	11:18:00	11:23:00	Member	S001	06C01	LOC06	Normal	56	21	5	Pending	Card
Q00015	27-06-2026	16:52:00	17:02:00	17:08:00	Member	S007	06C01	LOC06	High	18	10	6	No Show	Online
Q00016	13-05-2026	10:21:00	10:39:00	10:45:00	Senior Citizen	S007	03-C02	LOC03	Normal	69	18	6	Completed	Online
Q00017	07-05-2026	17:09:00	17:26:00	17:32:00	Senior Citizen	S005	01C02	LOC01	Normal	40	17	6	Completed	Online
Q00018	01-05-2026	14:43:00	14:51:00	14:59:00	Online	S008	02C02	LOC02	Normal	71	8	8	Completed	Online
Q00019	07-03-2026	13:17:37	13:25:00	13:34:00	Regular	S009	04C01	LOC04	Normal	38	8	9	Completed	N/A
Q00020	13-03-2026	16:33:00	16:33:00	16:39:00	Online	S007	04C03	LOC04	Normal	60	0	6	Completed	UPI
Q00021	29-05-2026	10:25:00	10:31:00	10:40:00	Member	S009	06C04	LOC06	Normal	45	6	9	Completed	N/A
Q00022	03-02-2026	10:59:00	11:07:00	11:10:00	Online	S001	04C05	LOC04	High	32	8	3	Completed	Card
Q00023	20-04-2026	11:31:00	11:43:00	11:56:00	Member	s009	04C05	LOC04	Normal	60	12	13	Completed	N/A
Q00024	12-03-2026	10:23:44	10:26:00	10:36:00	Online	S002	03C06	LOC03	Normal	60	3	10	No Show	Card
Q00025	23-02-2026	09:15:00	09:15:00	09:33:00	Senior Citizen	S003	03-C05	LOC03	Normal	53	0	18	Completed	Cash
Q00026	09-05-2026	17:23:00	17:31:00	17:36:00	Senior Citizen	S001	02C04	LOC02	High	23	8	5	Completed	Card
Q00027	07-03-2026	11:02:00	11:21:00	11:29:00	Senior Citizen	S008	01C05	LOC01	Normal	18	19	8	Completed	Online
Q00028	27-04-2026	12:51:00	13:00:00	Member	S009	03C02	LOC03	Normal	48	14	9	Completed	Cash	
Q00029	23-03-2026	09:23:00	09:32:00	09:41:00	Regular	S003	01C04	LOC01	Normal	19	9	9	Pending	Online
Q00030	22-04-2026	16:43:00	16:43:00	16:51:00	Online	S007	06C04	LOC06	Normal	47	0	8	Completed	UPI
Q00031	20-06-2026	09:57:00	10:10:00	10:21:00	Walk-in	S010	04C05	LOC04	Low	25	13	11	Completed	Online
Q00032	23-02-2026	10:33:00	10:44:00	10:52:00	Senior Citizen	S001	02C01	LOC02	Normal	67	11	8	Completed	N/A
Q00033	21-03-2026	12:59:00	13:12:00	13:15:00	Walk-in	S006	06C01	LOC06	Normal	27	13	3	Completed	Cash
Q00034	29-03-2026	10:07:00	10:19:00	10:30:00	Member	S009	06C03	LOC06	Normal	59	12	11	Completed	Online
Q00035	22-04-2026	10:59:00	11:11:00	11:16:00	Regular	S001	04-C05	LOC04	Normal	38	12	5	Completed	Card
Q00036	23-02-2026	10:16:00	10:22:00	10:26:00	Online	S003	03C04	LOC03	Urgent	71	6	4	Cancelled	UPI
Q00037	21-05-2026	17:08:00	17:27:00	17:50:00	Senior Citizen	S004	01C05	LOC01	Normal	19	19	23	Completed	n/a
Q00038	18-02-2026	11:27:00	11:36:00	11:44:00	Senior Citizen	S002	04C01	LOC04	Normal	66	9	8	Completed	Cash
Q00039	26-06-2026	14:30:00	14:32:00	14:44:00	Regular	S003	02C01	LOC02	NORMAL	42	2	12	Completed	Online
Q00040	02-05-2026	12:03:00	12:19:00	12:32:00	Regular	S005	01C04	loc01	Normal	35	16	13	Completed	Cash
Q00041	07-02-2026	10:45:00	10:56:00	11:03:00	Online	S007	03C04	LOC03	Normal	44	11	7	Completed	Card
Q00042	13-02-2026	16:41:00		17:06:00	Member	S003	01C06	LOC01	Normal	62	16	9	Completed	Online
Q00043	14-05-2026	12:03:00	12:08:00	12:16:00	Member	S002	05C05	LOC05	High	57	5	8	Completed	Cash

Project_Setup | Raw_Queue_Data | Cleaned_Data | Data_Validation | Module1_Analysis | Service_Master | Location_Mast

Figure 7.3: *Cleaned Queue Data – Columns A to O*

P	Q	R	S	T	U	V	W	X
Service_Fee	Staff_ID	Satisfaction_Score	Complaint	Remarks	Service_Name	Location_Name	Wait_Category	Priority_Risk
250	STF0203	5	None	Priority case	Registration	Market Branch	Moderate Wait	Normal
100	STF0106	4	None	Needs review	Cash Deposit	Central Branch	Moderate Wait	Normal
0	STF0502	5	None	0	General Enquiry	Main Branch	Low Wait	Normal
0	STF0101	5	None	Priority case	Cash Withdrawal	Central Branch	Moderate Wait	Normal
250	STF0101	5	None	Priority case	Registration	Central Branch	Low Wait	Normal
250	STF0203	3	None	Verified	Registration	Market Branch	Low Wait	Normal
100	STF0504	5	None	Verified	Bill Payment	Main Branch	Moderate Wait	Normal
100	STF0304	4	Payment Issue	Verified	Bill Payment	Station Branch	Low Wait	Attention Required
100	STF0603	4	None	Needs review	Cash Withdrawal	East Branch	Low Wait	Normal
150	STF0601	4	None	Needs review	Account Service	East Branch		Attention Required
0	STF0604	5	None	Priority case	Complaint	East Branch	Low Wait	Normal
100	STF0402	1	None	0	Cash Deposit	City Branch	Moderate Wait	Normal
200	STF0106	3	None	Priority case	Document Verification	Central Branch	Low Wait	Normal
100	STF0601	1	Service Issue	Verified	Bill Payment	East Branch	High Wait	Attention Required
100	STF0601	2	None	Follow-up	Cash Deposit	East Branch	Low Wait	Normal
100	STF0302	2	Payment Issue	Follow-up	Cash Deposit	Station Branch	Moderate Wait	Attention Required
0	STF0102	2	None	Verified	Complaint	Central Branch	Moderate Wait	Normal
100	STF0202	5	None	0	Cash Withdrawal	Market Branch	Low Wait	Normal
150	STF0401	1	None	Needs review	Application Submission	City Branch	Low Wait	Normal
100	STF0403	1	None	0	Cash Deposit	City Branch	Low Wait	Normal
150	STF0604	1	None	Verified	Application Submission	East Branch	Low Wait	Normal
100	STF0405	5	None	Needs review	Bill Payment	City Branch	Low Wait	Normal
0	STF0405	3	None	Verified	Application Submission	City Branch	Moderate Wait	Normal
150	STF0306	2	None	Needs review	Account Service	Station Branch	Low Wait	Normal
200	STF0305	4	None	Verified	Document Verification	Station Branch	Low Wait	Normal
100	STF0204	1	None	Priority case	Bill Payment	Market Branch	Low Wait	Normal
100	STF0105	4	None	Needs review	Cash Withdrawal	Central Branch	Moderate Wait	Normal
0	STF0302	3	None	Follow-up	Application Submission	Station Branch	Moderate Wait	Normal
200	STF0104	1	Payment Issue	Follow-up	Document Verification	Central Branch	Low Wait	Attention Required
100	STF0604	3	None	Needs review	Cash Deposit	East Branch	Low Wait	Normal
200	STF0405	3	None	Verified	Certificate Request	City Branch	Moderate Wait	Normal
100	STF0201	3	None	Needs review	Bill Payment	Market Branch	Moderate Wait	Normal
0	STF0601	2	None	Follow-up	General Enquiry	East Branch	Moderate Wait	Normal
0	STF0603	1	Service Issue	0	Application Submission	East Branch	Moderate Wait	Attention Required
0	STF0405	2	Staff Issue	0	Bill Payment	City Branch	Moderate Wait	Attention Required
200	STF0304	4	None	Verified	Document Verification	Station Branch	Low Wait	Normal
250	STF0105	1	Service Issue	Follow-up	Registration	Central Branch	Moderate Wait	Attention Required
250	STF0401	5	Long Wait	Verified	Account Service	City Branch	Low Wait	Attention Required
200	STF0201	5	None	Follow-up	Document Verification	Market Branch	Low Wait	Normal
0	STF0104	1	None	0	Complaint	Central Branch	Moderate Wait	Normal
100	STF0304	5	None	Needs review	Cash Deposit	Station Branch	Moderate Wait	Normal
200	STF0106	5	Long Wait	Priority case	Document Verification	Central Branch	Moderate Wait	Attention Required
150	STF0505	1	None	Follow-up	Account Service	Main Branch	Low Wait	Normal

Figure 7.4: *Cleaned Queue Data – Columns P to X*

7.3 Data Validation Results

The cleaned dataset was validated using Excel formulas.

Validation Check	Result
Total Queue Records	10,000
Total Missing Values	469
Duplicate Queue ID Values	86
Negative Waiting Time	49
Negative Service Time	39
Invalid Customer Age	0
Invalid Satisfaction Score	90
Negative Service Fee	0
Zero Service Time	109
Zero Waiting Time	956

These checks helped identify data-quality issues before further analysis.

Field	Missing Values
Queue_ID	0
Date	0
Arrival_Time	0
Service_Start_Time	276
Service_End_Time	193
Customer_Type	0
Service_ID	0
Counter_ID	0
Location_ID	0
Priority	0
Customer_Age	0
Waiting_Time	49
Service_Time	39
Status	0
Payment_Mode	0
Service_Fee	0
Staff_ID	0
Satisfaction	90
Complaint_Flag	0
Remarks	0

Figure 7.5: *Missing Value Validation Results*

Total Queue Records
Formula: *=COUNTA(Cleaned_Data!A2:A10001)*

Total Missing Values
Formula: *=COUNTBLANK(Cleaned_Data!B2:B10001)*

Duplicate Queue ID
Formula: `=COUNTIF(B26:B10025,">1")`

Negative Waiting Time
Formula: `=COUNTIF(Cleaned_Data!L2:L10001,"<0")`

Negative Service Time
Formula: `=COUNTIF(Cleaned_Data!M2:M10001,"<0")`

Invalid Satisfaction
Formula:
=COUNTIF(Cleaned_Data!R2:R10001,"<1")+COUNTIF(Cleaned_Data!R2:R10001,">5")

Zero Service Time
Formula: *=COUNTIF(Cleaned_Data!M2:M10001,"=0")*

Zero Waiting Time
Formula: *=COUNTIF(Cleaned_Data!L2:L10001,"=0")*

7.4 Basic Queue Analysis

The following key metrics were calculated using Excel formulas:

Metric	Result
Total Queue Records	10,000
Average Waiting Time	10.46 minutes
Maximum Waiting Time	36 minutes
Minimum Waiting Time	0 minutes
Critical Wait Cases	16
Urgent Customers	852
Complaint Cases	1,810
Average Satisfaction	2.94

These metrics provide an overview of queue performance and customer experience.

QueueFlow Module 1 Analysis	
Urgent Customers Total Wait	8718
Total Queue Records	10000
Average Waiting Time	10.46316953
Maximum Waiting Time	36
Minimum Waiting Time	0
Critical Wait Cases	16
Urgent Customers	852
Complaint Cases	1810
Average Satisfaction	2.943794147

Figure 7.6: *Queue Performance Summary*

Total Queue Records
Formula: *=COUNTA(Cleaned_Data!A2:A10001)*

Average Waiting Time
Formula: *=AVERAGE(Cleaned_Data!L2:L10001)*

Maximum Waiting Time
Formula: *=MAX(Cleaned_Data!L2:L10001)*

Minimum Waiting Time
Formula: *=MIN(Cleaned_Data!L2:L10001)*

Critical Wait Cases
Formula: *=COUNTIF(Cleaned_Data!L2:L10001,">30")*

Urgent Customers
Formula: *=COUNTIF(Cleaned_Data!J2:J10001,"Urgent")*

Complaint Cases
Formula*: =COUNTIF(Cleaned_Data!S2:S10001,"Yes")*

Average Satisfaction
Formula: *=AVERAGE(Cleaned_Data!R2:R10001)*

7.5 Dynamic Array Analysis

Dynamic array functions were used to analyse the cleaned dataset.

`UNIQUE` was used to identify distinct customer categories. `FILTER` was used to extract records based on waiting time, while `SORT` and `SORTBY` were used to arrange records.

Example:

```
=SORTBY(FILTER(Cleaned_Data!A2:X10001,Cleaned_Data!L2:L10001>20
),FILTER(Cleaned_Data!L2:L10001,Cleaned_Data!L2:L10001>20),-1)
```

This helped identify records with longer waiting times.

Dynamic Array & Conditional Filtering Analysis																			
Q00010	46104	0.736805556	0.736805556	0.743055556	Walk-in	S002	06C01	LOC06	Normal	73	9	Completed	N/A	150	STF0601	4	None	Needs review
Q00065	46169	0.398831019	0.409027778	0.411805556	Regular	S001	04C01	loc04	Normal	25	4	Completed	Cash	100	STF0401	5	None	Follow-up
Q00091	46172	0.388888889	0.39375	0.402083333	Regular	S004	01C06	LOC01	High	40	12	Completed	Online	250	STF0106	2	None	Needs review
Q00427	46081	0.654861111	0.676388889	0.679861111	Regular	S005	06C02	LOC06	Normal	32	31	5	Completed	N/A	0	STF0602	5	None	0
Q00562	46179	0.421527778	0.443055556	0.453472222	Regular	S004	04C04	LOC04	High	21	31	15	Completed	Card	250	STF0404	1	None	Priority case
Q00867	46079	0.688888889	0.711111111	0.715972222	Regular	S007	06C04	LOC06	Urgent	18	32	7	Completed	Online	100	STF0604	1	Staff Issue	Verified
Q01252	46194	0.563194444	0.572222222	0.579166667	Senior Citizen	S005	01C05	LOC01	NORMAL	59	10	Completed	Card	0	STF0105	3	None	Priority case
Q01469	46157	0.690972222	0.7125	0.716666667	Online	S007	01C02	LOC01	Normal	58	31	6	Completed	Online	100	STF0102	2	Service Issue	Follow-up
Q01489	46141	0.583333333	0.590972222	0.597916667	Member	S003	01C04	LOC01	Normal	21	10	Completed	UPI	200	STF0104	4	None	Verified
Q01491	46143	0.555555556	0.563194444	0.565277778	Member	S006	03C03	LOC03	Normal	59	3	Completed	N/A	0	STF0303	4	None	Follow-up
Q01985	46158	0.475	0.495833333	0.500694444	Member	S008	05C01	LOC05	Normal	61	7	Completed	N/A	100	STF0501	1	None	Needs review
Q02030	46130	0.392361111		0.403472222	Online	S007	02-C05	LOC02	Normal	50	5	Completed	N/A	100	STF0205		None	0
Q02517	46133	0.381944444	0.388888889	0.393055556	Member	S004	04C03	LOC04	Normal	61	6	Completed	N/A	250	STF0403	2	None	Follow-up
Q02531	46062	0.604861111	0.613194444	0.619444444	Regular	S008	06C02	LOC06	Normal	38	9	Completed	N/A	0	STF0602	2	None	Follow-up
Q02567	46170	0.5	0.507638889	0.5125	Online	S002	06C04	LOC06	High	57	7	Completed	UPI	150	STF0604	3	None	Follow-up
Q02580	46041	0.535416667	0.556944444	0.561805556	Member	S007	01-C02	LOC01	Urgent	71	31	7	Completed	Card	100	STF0102	1	None	Verified
Q02592	46069	0.557638889	0.579861111	0.589583333	Walk-in	S003	01C06	LOC01	Normal	38	32	14	Completed	Online	200	STF0106	1	None	0
Q02614	46072	0.713842593	0.717361111	0.721527778	Member	S007	05C01	LOC05	Normal	35	6	Completed	N/A	125	STF0501	4	None	0
Q02816	46197	0.622222222	0.629166667	0.636805556	Member	S009	03C02	LOC03	Normal	56	11	Completed	UPI	0	STF0302	4	None	Follow-up
Q03149	46165	0.490277778	0.499305556	0.505555556	Online	S002	03C03	LOC03	Urgent	73	9	cancelled	UPI	150	STF0303	1	None	Needs review
Q03941	46172	0.6625	0.685416667	0.6875	Online	S006	05C04	loc05	Normal	39	33	3	Completed	Card	0	STF0504	2	Service Issue	0
Q03977	46141	0.413888889	0.434027778	0.445138889	Member	S004	03C05	LOC03	Normal	71	16	Completed	N/A	250	STF0305	2	None	Verified
Q04014	46119	0.597222222	0.599305556	0.60625	Senior Citizen	S003	03C02	LOC03	High	48	10	Completed	Card	300	STF0302	1	None	0
Q04163	46176	0.526388889	0.536111111	0.546527778	Online	S004	05C02	LOC05	URGENT	52	15	Completed	Online	250	STF0502	1	Staff Issue	Priority case
Q04295	46183	0.625	0.634027778	0.640972222	Regular	S002	03C03	LOC03	Normal	69	10	No Show	Card	175	STF0303	3	Long Wait	Needs review
Q04444	46141	0.391666667	0.391666667	0.397222222	Senior Citizen	S004	06C01	LOC06	High	59	8	Completed	Cash	250	STF0601	5	None	Verified
Q04736	46164	0.38125	0.392361111	0.398611111	Member	S004	02C05	LOC02	Urgent	51	9	Completed	N/A	275	STF0205	5	None	Verified
Q04750	46115	0.414583333	0.427777778	0.438888889	Walk-in	S010	01C02	LOC01	Normal	46	16	No Show	Card	200	STF0102	2	None	0
Q04757	46183	0.727777778	0.7375	0.740277778	Walk-in	S001	04C01	LOC04	Normal	41	4	Completed	UPI	100	STF0401	2	None	Priority case
Q04764	46058	0.671527778	0.673611111	0.677777778	Member	S008	04-C03	LOC04	Normal	36	6	Completed	Card	100	STF0403	3	None	Follow-up
Q04861	46027	0.708333333	0.709722222	0.711805556	Walk-in	S001	01C04	LOC01	High	57	3	Completed	UPI	100	STF0104	4	None	Verified
Q04884	46189	0.425601852	0.436111111	0.440277778	Senior Citizen	S001	01C05	LOC01	Normal	30	6	Completed	N/A	100	STF0105	5	None	Needs review
Q05073	46188	0.566666667	0.589583333	0.597916667	Walk-in	S003	02C04	LOC02	Normal	70	33	12	Completed	UPI	200	STF0204	5	None	Follow-up
Q05108	46173	0.740972222	0.752777778	0.760416667	Walk-in	S010	03C06	LOC03	High	36	11	No Show	Online	200	STF0306	2	None	Verified
Q05175	46134	0.429166667	0.438194444	0.447916667	Senior Citizen	S010	04C03	LOC04	High	53	14	Completed	N/A	200	STF0403	3	None	Needs review
Q05190	46063	0.484027778	0.49375	0.504166667	Member	S010	04C05	LOC04	Normal	57	15	Completed	Cash	200	STF0405	1	None	Verified
Q05290	46098	0.546527778	0.56875	0.575	Regular	S009	02C02	LOC02	Normal	55	32	9	Completed	Online	150	STF0202	2	None	Needs review
Q05369	46169	0.422222222	0.445833333	0.450694444	Senior Citizen	S007	02C03	LOC02	Urgent	49	34	7	Completed	N/A	100	STF0203	3	Long Wait	Priority case
Q05889	46070	0.659722222	0.663888889		Member	S004	01C05	LOC01	Normal	37	15	Completed	Cash	250	STF0105	4	None	Needs review
Q05949	46029	0.622916667	0.638194444	0.645138889	Member	S005	02C01	LOC02	Normal	62		Completed	Online	0	STF0201	1	None	Priority case
Q06093	46129	0.714583333	0.719444444	0.722222222	Online	S006	02C05	loc02	Normal	39	4	Completed	Online	0	STF0205	4	Payment Issue	Verified
Q06288	46125	0.567361111		0.582638889	Member	S008	04C01	LOC04	Normal	69	7	Completed	Online	0	STF0401	4	Service Issue	0
Q06319	46122	0.405555556	0.429861111		Online	S003	01C06	LOC01	Normal	28	35	18	Completed	Online	200	STF0106	2	None	Verified
Q06799	46137	0.484027778	0.495833333	0.502083333	Walk-in	S008	03C01	LOC03	Normal	66	9	Pending	Cash	100	STF0301	4	None	0
Q06861	46166	0.521099537	0.53125	0.5375	Member	S009	02C01	LOC02	NORMAL	46	9	No Show	n/a	150	STF0201	5	Long Wait	Follow-up
Q06866	46130	0.543055556	0.545138889	0.551388889	Regular	S009	05C04	LOC05	Urgent	62	9	Completed	Cash	150	STF0504	2	None	Verified
Q07071	46038	0.631944444	0.636111111	0.640277778	Senior Citizen	S008	02C03	LOC02	High	38	6	Completed	UPI	100	STF0203	3	None	Verified

Figure 7.7: *Dynamic Array and Conditional Filtering Results*

UNIQUE – Customer Types
Formula: `=UNIQUE(Cleaned_Data!F2:F10001)`

SORT – Customer Types
Formula: `=SORT(UNIQUE(Cleaned_Data!F2:F10001))`

FILTER – Waiting Time > 20
Formula: `=FILTER(Cleaned_Data!A2:X10001,Cleaned_Data!L2:L10001>20,"No records found")`

SORTBY – Waiting Time Descending
Formula:
`=SORTBY(FILTER(Cleaned_Data!A2:X10001,Cleaned_Data!L2:L10001>20),FILTER(Cleaned_Data!L2:L10001,Cleaned_Data!L2:L10001>20),-1)`

7.6 Lookup and Reference Analysis

Lookup functions were used to connect queue records with master data.
These functions helped connect the queue data with Service and Location master tables.

INDEX + MATCH Lookup	
Enter Service ID	S001
Service Name	Bill Payment

XLOOKUP Lookup	
Enter Service ID	S005
Service Name	Complaint

Figure 7.8: *INDEX + MATCH and XLOOKUP Analysis*

INDEX + MATCH – Service Name
Formula: =INDEX(Service_Master!B:B,MATCH(E4,Service_Master!A:A,0))

XLOOKUP – Service Name
Formula: =XLOOKUP(E8,Service_Master!A:A,Service_Master!B:B,"Not Found")

Location Lookup – Location Name
Formula: =INDEX(Location_Master!B:B,MATCH(A32,Location_Master!A:A,0))

XLOOKUP – Location Name
Formula: =XLOOKUP(A32,Location_Master!A:A,Location_Master!B:B,"Not Found")

7.7 Service-wise Waiting Time Analysis

Average waiting time was calculated for each service using `AVERAGEIF`.

Service ID	Service Name	Average Waiting Time
S001	Bill Payment	10.70
S002	Account Service	10.59
S003	Document Verification	10.63
S004	Registration	10.66
S005	Complaint	10.67
S006	General Enquiry	10.65
S007	Cash Deposit	10.14
S008	Cash Withdrawal	9.91
S009	Application Submission	10.58
S010	Certificate Request	10.09

This analysis helps compare waiting time across different services.

S001 – Bill Payment
Formula: `=AVERAGEIF(Cleaned_Data!G2:G10001,"S001",Cleaned_Data!L2:L10001)`

S002 – Account Service
Formula: `=AVERAGEIF(Cleaned_Data!G2:G10001,"S002",Cleaned_Data!L2:L10001)`

S003 – Document Verification
Formula: `=AVERAGEIF(Cleaned_Data!G2:G10001,"S003",Cleaned_Data!L2:L10001)`

S004 – Registration
Formula: `=AVERAGEIF(Cleaned_Data!G2:G10001,"S004",Cleaned_Data!L2:L10001)`

S005 – Complaint
Formula: `=AVERAGEIF(Cleaned_Data!G2:G10001,"S005",Cleaned_Data!L2:L10001)`

S006 – General Enquiry
Formula: `=AVERAGEIF(Cleaned_Data!G2:G10001,"S006",Cleaned_Data!L2:L10001)`

S007 – Cash Deposit
Formula: `=AVERAGEIF(Cleaned_Data!G2:G10001,"S007",Cleaned_Data!L2:L10001)`

S008 – Cash Withdrawal
Formula: `=AVERAGEIF(Cleaned_Data!G2:G10001,"S008",Cleaned_Data!L2:L10001)`

S009 – Application Submission
Formula: `=AVERAGEIF(Cleaned_Data!G2:G10001,"S009",Cleaned_Data!L2:L10001)`

S010 – Certificate Request
Formula: `=AVERAGEIF(Cleaned_Data!G2:G10001,"S010",Cleaned_Data!L2:L10001)`

7.8 Customer Type Analysis

Average waiting time was analysed according to customer type.

Customer Type	Average Waiting Time
Regular	10.53
Member	10.36
Senior Citizen	10.49
Walk-in	10.20
Online	10.64

This provides a comparison of queue conditions for different customer groups.

Customer Type	Average Waiting Time
Regular	10.52925532
Member	10.35942327
Senior Citizen	10.49272349
Walk-in	10.20372233
Online	10.64229637

Figure 7.9: *Customer Type-wise Waiting Time Analysis*

Regular – Average Waiting Time
Formula: `=AVERAGEIF(Cleaned_Data!F2:F10001,"Regular",Cleaned_Data!L2:L10001)`

Member – Average Waiting Time
Formula: `=AVERAGEIF(Cleaned_Data!F2:F10001,"Member",Cleaned_Data!L2:L10001)`

Senior Citizen – Average Waiting Time
Formula: `=AVERAGEIF(Cleaned_Data!F2:F10001,"Senior Citizen",Cleaned_Data!L2:L10001)`

Walk-in – Average Waiting Time
Formula: `=AVERAGEIF(Cleaned_Data!F2:F10001,"Walk-in",Cleaned_Data!L2:L10001)`

Online – Average Waiting Time
Formula: `=AVERAGEIF(Cleaned_Data!F2:F10001,"Online",Cleaned_Data!L2:L10001)`

7.9 Priority-wise Analysis

Priority-level analysis was performed using `AVERAGEIF` and `COUNTIF`.

Priority	Average Waiting Time	Queue Count
Normal	10.51	6,974
Low	10.50	736
High	10.13	1,438
Urgent	10.29	852

The Queue Count values total 10,000.

Priority	Average Waiting Time	Queue Count
Normal	10.51073642	6974
Low	10.50271739	736
High	10.31280616	1438
Urgent	10.29279811	852

Figure 7.10: *Priority-wise Queue Analysis*

Normal – Average Waiting Time
Formula: `=AVERAGEIF(Cleaned_Data!J2:J10001,"Normal",Cleaned_Data!L2:L10001)`

Low – Average Waiting Time
Formula: `=AVERAGEIF(Cleaned_Data!J2:J10001,"Low",Cleaned_Data!L2:L10001)`

High – Average Waiting Time
Formula: `=AVERAGEIF(Cleaned_Data!J2:J10001,"High",Cleaned_Data!L2:L10001)`

Urgent – Average Waiting Time
Formula: `=AVERAGEIF(Cleaned_Data!J2:J10001,"Urgent",Cleaned_Data!L2:L10001)`

Normal – Queue Count
Formula: `=COUNTIF(Cleaned_Data!J2:J10001,"Normal")`

Low – Queue Count
Formula: `=COUNTIF(Cleaned_Data!J2:J10001,"Low")`

High – Queue Count
Formula: `=COUNTIF(Cleaned_Data!J2:J10001,"High")`

Urgent – Queue Count
Formula: `=COUNTIF(Cleaned_Data!J2:J10001,"Urgent")`

7.10 Location-wise Analysis

Location-wise waiting time and queue volume were calculated using `AVERAGEIF` and `COUNTIF`.

Location ID	Location Name	Average Waiting Time	Queue Count
LOC01	Central Branch	10.31	1,685
LOC02	Market Branch	10.48	1,677
LOC03	Station Branch	10.39	1,629
LOC04	City Branch	10.43	1,681
LOC05	Main Branch	10.57	1,667
LOC06	East Branch	10.49	1,661

This provides a branch-level view of queue performance.

Location ID	Location Name	Average Waiting Time	Queue Count
LOC01	Central Branch	10.31246273	1685
LOC02	Market Branch	10.48292391	1677
LOC03	Station Branch	10.39851943	1629
LOC04	City Branch	10.43345324	1681
LOC05	Main Branch	10.65963855	1667
LOC06	East Branch	10.49214976	1661

Figure 7.10: *Location-wise Queue Analysis*

LOC01 – Average Waiting Time
Formula: `=AVERAGEIF(Cleaned_Data!I2:I10001,"LOC01",Cleaned_Data!L2:L10001)`

LOC02 – Average Waiting Time
Formula: `=AVERAGEIF(Cleaned_Data!I2:I10001,"LOC02",Cleaned_Data!L2:L10001)`

LOC03 – Average Waiting Time
Formula: `=AVERAGEIF(Cleaned_Data!I2:I10001,"LOC03",Cleaned_Data!L2:L10001)`

LOC04 – Average Waiting Time
Formula: `=AVERAGEIF(Cleaned_Data!I2:I10001,"LOC04",Cleaned_Data!L2:L10001)`

LOC05 – Average Waiting Time
Formula: `=AVERAGEIF(Cleaned_Data!I2:I10001,"LOC05",Cleaned_Data!L2:L10001)`

LOC06 – Average Waiting Time
Formula: `=AVERAGEIF(Cleaned_Data!I2:I10001,"LOC06",Cleaned_Data!L2:L10001)`

LOC01 – Queue Count
Formula: `=COUNTIF(Cleaned_Data!I2:I10001,"LOC01")`

LOC02 – Queue Count
Formula: `=COUNTIF(Cleaned_Data!I2:I10001,"LOC02")`

LOC03 – Queue Count
Formula: `=COUNTIF(Cleaned_Data!I2:I10001,"LOC03")`

LOC04 – Queue Count
Formula: `=COUNTIF(Cleaned_Data!I2:I10001,"LOC04")`

LOC05 – Queue Count
Formula: `=COUNTIF(Cleaned_Data!I2:I10001,"LOC05")`

LOC06 – Queue Count
Formula: `=COUNTIF(Cleaned_Data!I2:I10001,"LOC06")`

7.11 Logical Classification

Logical functions were used to create derived fields such as **Wait Category** and **Priority Risk**.

Wait Category includes:

- Low Wait
- Moderate Wait
- High Wait

Priority Risk identifies records requiring additional attention based on queue conditions.

These classifications make the data more useful for further analysis and dashboard development.

Wait_Category	Priority_Risk
Moderate Wait	Normal
Moderate Wait	Normal
Low Wait	Normal
Moderate Wait	Normal
Low Wait	Normal
Low Wait	Normal
Moderate Wait	Normal
Low Wait	Attention Required
Low Wait	Normal
	Attention Required
Low Wait	Normal
Moderate Wait	Normal
Low Wait	Normal
High Wait	Attention Required
Low Wait	Normal
Moderate Wait	Attention Required
Moderate Wait	Normal
Low Wait	Normal
Low Wait	Normal
Low Wait	Normal
Low Wait	Normal
Low Wait	Normal
Moderate Wait	Normal
Low Wait	Normal
Low Wait	Normal
Low Wait	Normal
Moderate Wait	Normal
Moderate Wait	Normal
Low Wait	Attention Required
Low Wait	Normal
Moderate Wait	Normal
Moderate Wait	Normal
Moderate Wait	Normal
Moderate Wait	Attention Required
Moderate Wait	Attention Required
Low Wait	Normal
Moderate Wait	Attention Required
Low Wait	Attention Required
Low Wait	Normal
Moderate Wait	Normal
Moderate Wait	Normal
Moderate Wait	Attention Required
Low Wait	Normal

Figure 7.11: *Wait Category and Priority Risk*

Wait Category
Formula:

```
=IF(L2<=10,"Low Wait",IF(L2<=20,"Moderate Wait","High Wait"))
```

Priority Risk
Formula:

```
=IF(AND(J2="Urgent",L2>20),"Attention
Required",IF(OR(L2>30,S2="Yes"),"Attention Required","Normal"))
```

S

7.12 Module 1 Outcome

Module 1 transformed the raw QueueFlow dataset into a structured and analysis-ready dataset. Data-quality issues were identified, values were standardized, and Excel functions were used for cleaning, lookup, classification and analysis.

The module produced useful views of queue performance based on **customer type, service, priority and location**. The cleaned dataset will be used as the foundation for **Module 2 – Analytics & Forecasting**.

MODULE 2 – ANALYTICS & FORECASTING

8.1 Module Overview

Module 2 of the QueueFlow Analytics project focuses on analysing the cleaned queue dataset using advanced Excel analytics, forecasting, date and time functions, and financial decision-making techniques.

The cleaned dataset contains **10,000 queue records** covering the period from **01 January 2026 to 29 June 2026**. Daily and monthly queue patterns were analysed using queue count and average waiting time.

Excel functions such as **AVERAGEIF, AVERAGEIFS, COUNTIF, COUNTIFS, COUNTIFS, FORECAST, TREND, RSQ, LINEST, EOMONTH, WORKDAY, NETWORKDAYS, DATEDIF, NPV, IRR, PMT and Goal Seek** were used for analytics, forecasting and decision-making.

8.2 Time-Based Queue Analysis

The cleaned dataset was analysed to determine the overall operational period and queue performance.

Metric	Result
Earliest Queue Date	01-01-2026
Latest Queue Date	29-06-2026
Total Days Covered	180
Working Days Covered	128
Average Waiting Time	10.46 minutes
Total Queue Records	10,000

Earliest Queue Date
Formula: `=MIN(Cleaned_Data!B2:B10001)`

Latest Queue Date
Formula: `=MAX(Cleaned_Data!B2:B10001)`

Total Days Covered
Formula: `=B3-B2+1`

Working Days Covered
Formula: `=NETWORKDAYS(B2,B3)`

Average Waiting Time
Formula: `=AVERAGE(Cleaned_Data!L2:L10001)`

Total Queue Records
Formula: `=COUNTA(Cleaned_Data!A2:A10001)`

QueueFlow Module 2 – Analytics & Forecasting	
Earliest Queue Date	01-01-2026
Latest Queue Date	29-06-2026
Total Days Covered	180
Working Days Covered	128
Average Waiting Time	10.46316953
Total Queue Records	10000

Figure 8.1: Time-Based Queue Analysis

8.3 Daily Queue Analysis

Daily queue counts and average waiting times were calculated for each date in the dataset. This analysis helps identify variations in queue volume and waiting-time behaviour throughout the observation period.

Daily Queue Count

Formula: *=COUNTIF(Cleaned_Data!B2:B10001,A12)*

Daily Average Waiting Time

Formula: *=AVERAGEIF(Cleaned_Data!B2:B10001,A12,Cleaned_Data!L2:L10001)*

Date	Queue Count	Average Waiting Time
01-01-2026	56	10.57142857
02-01-2026	46	10.47826087
03-01-2026	63	10.3015873
04-01-2026	67	11.53731343
05-01-2026	67	10.671875
06-01-2026	56	10.60714286
07-01-2026	46	11.37777778
08-01-2026	50	10.84
09-01-2026	62	10.74193548
10-01-2026	53	9.245283019
11-01-2026	52	10.61538462
12-01-2026	53	10.43396226
13-01-2026	62	9.14516129
14-01-2026	67	10.91044776
15-01-2026	54	10.77777778
16-01-2026	49	11
17-01-2026	59	11.44067797
18-01-2026	47	12
19-01-2026	47	10.80851064
20-01-2026	42	12.4047619
21-01-2026	49	9.673469388
22-01-2026	44	8.139534884
23-01-2026	53	9.622641509
24-01-2026	56	10.53571429
25-01-2026	44	10.79545455
26-01-2026	58	11.15517241
27-01-2026	72	11.26388889
28-01-2026	52	10.78846154
29-01-2026	67	10.44776119
30-01-2026	57	9.947368421
31-01-2026	50	11.06
01-02-2026	60	10.46666667
02-02-2026	56	11.76785714
03-02-2026	48	9.936170213
04-02-2026	48	9.875
05-02-2026	59	9.293103448
06-02-2026	60	10.78333333
07-02-2026	43	10.6744186
08-02-2026	44	9.545454545
09-02-2026	47	9.891304348
10-02-2026	57	10.64285714
11-02-2026	55	10.38181818
12-02-2026	56	9.767857143
13-02-2026	63	10.15873016
14-02-2026	55	9.909090909
15-02-2026	63	11.41269841
16-02-2026	60	11.43333333

Figure 8.2: Daily Queue Count and Average Waiting Time Analysis

8.4 Monthly Queue Analysis

The daily queue information was aggregated into monthly values to understand broader queue patterns.

Month	Queue Count	Average Waiting Time
January 2026	1,700	10.6269
February 2026	1,506	10.3189
March 2026	1,762	10.5270

April 2026	1,687	10.4626
May 2026	1,764	10.4083
June 2026	1,581	10.4146

Monthly Queue Count
Formula:
```
=COUNTIFS(Cleaned_Data!B$2:B$10001,">="&E12,Cleaned_Data!B$2:B$10001
,"<"&EOMONTH(E12,0)+1)
```

Monthly Average Waiting Time
Formula:
```
=AVERAGEIFS(Cleaned_Data!L$2:L$10001,Cleaned_Data!B$2:B$10001,">="&E
12,Cleaned_Data!B$2:B$10001,"<"&EOMONTH(E12,0)+1)
```

Month	Queue Count	Average Waiting Time
01-01-2026	1700	10.62691854
01-02-2026	1506	10.31887925
01-03-2026	1762	10.52700398
01-04-2026	1687	10.46264196
01-05-2026	1764	10.40833809
01-06-2026	1581	10.41460317

Figure 8.3: Monthly Queue Analysis

8.5 Forecasting Analysis

Historical monthly average waiting-time values were used to estimate the expected waiting time for the next period.

The **FORECAST, TREND, RSQ and LINEST** functions were used for forecasting and trend analysis.

The forecasted average waiting time was approximately **10.37 minutes**.

FORECAST – July Waiting Time
Formula: `=FORECAST(E27,$G$12:$G$17,$E$12:$E$17)`
Result: **10.37210871 minutes**

TREND Forecast
Formula: `=TREND($G$12:$G$17,$H$12:$H$17,7)`
Result: **10.3739746 minutes**

Forecasting Analysis		
Month	**Actual Average Waiting Time**	**Forecasted Waiting Time**
01-01-2026	10.62691854	
01-02-2026	10.31887925	
01-03-2026	10.52700398	
01-04-2026	10.46264196	
01-05-2026	10.40833809	
01-06-2026	10.41460317	
01-07-2026		10.37210871
	TREND Forecast	10.3739746
	RSQ	0.184305119
	LINEST Regression Slope	-0.02450178

Figure 8.4: Forecasting Analysis

8.6 RSQ and LINEST Analysis

Statistical functions were used to evaluate the relationship and trend between the monthly sequence and average waiting time.

RSQ
Formula: `=RSQ($G$12:$G$17,$H$12:$H$17)`
Result: **0.184305119**

LINEST Regression Slope
Formula: `=INDEX(LINEST($G$12:$G$17,$H$12:$H$17),1,1)`
Result: **-0.02450178**

These calculations provide additional statistical support for the forecasting analysis.

RSQ	0.184305119
LINEST Regression Slope	-0.02450178

Figure 8.5: RSQ and LINEST Analysis

8.7 Date and Working-Day Analysis

Date and time functions were used to analyse the operational period represented by the queue dataset.

Next Working Date
Formula: `=WORKDAY(B3,5)`

Working Days
Formula: `=NETWORKDAYS(B2,B3)`

Date Difference – Days
Formula: `=DATEDIF(B2,B3,"d")`
Result: **179 days**

Date Difference – Months
Formula: `=DATEDIF(B2,B3,"m")`
Result: **5 months**

Next Working Date	46209
Working Days Check	128
Date Difference (Days)	179
Date Difference (Months)	5

Figure 8.6: Date and Working-Day Analysis

8.8 Operational Queue Analysis

Additional conditional analyses were performed using queue priority, status and service-fee information.

Analysis	Result
Urgent + Completed Average Waiting Time	10.4204 min
Urgent + Completed Queue Count	727
High Priority Average Waiting Time	10.3128 min
High Priority + Completed Average Waiting Time	10.3208 min
High Priority Queue Count	1,438
Average Monthly Queue Count	1,666.67
Completed Average Waiting Time	10.4716 min
Completed Average Service Fee	₹117.74

Urgent + Completed – Average Waiting Time
Formula:
```
=AVERAGEIFS(Cleaned_Data!L2:L10001,Cleaned_Data!J2:J10001,"Urgent",Cleaned_Data!N2:N10001,"Completed")
```

Urgent + Completed – Queue Count
Formula:
```
=COUNTIFS(Cleaned_Data!J2:J10001,"Urgent",Cleaned_Data!N2:N10001,"Completed")
```

High Priority – Average Waiting Time
Formula:
```
=AVERAGEIFS(Cleaned_Data!L2:L10001,Cleaned_Data!J2:J10001,"High")
```

High Priority + Completed – Average Waiting Time
Formula:
```
=AVERAGEIFS(Cleaned_Data!L2:L10001,Cleaned_Data!J2:J10001,"High",Cleaned_Data!N2:N10001,"Completed")
```

Average Monthly Queue Count
Formula: `=AVERAGE(F12:F17)`

Completed – Average Waiting Time
Formula:
```
=AVERAGEIF(Cleaned_Data!N2:N10001,"Completed",Cleaned_Data!L2:L10001)
```

Completed – Total Service Revenue
Formula:
```
=SUMIF(Cleaned_Data!N2:N10001,"Completed",Cleaned_Data!P2:P10001)
```

Completed – Average Service Fee
Formula:
```
=AVERAGEIF(Cleaned_Data!N2:N10001,"Completed",Cleaned_Data!P2:P10001)
```

Urgent Completed – Average Waiting Time	10.42047026
Urgent Completed – Queue Count	727
Main Branch – High Priority Average Waiting Time	9.770072993
Average Monthly Queue Count	1666.666667
High Priority Queue Count	1438
High Priority – Average Waiting Time	10.31280616
High Priority – Completed Average Waiting Time	10.32076985
Next Step — Completed Customers ka Average Waiting Time	10.47161674
Completed – Average Service Fee	117.7386614

Figure 8.7: Operational Queue Analysis

8.9 Financial Decision Model – Additional Counter

A financial decision model was created to analyse the financial requirements of introducing an additional service counter.

Parameter	Value
Initial Counter Investment	₹50,000
Monthly Operating Cost	₹8,000
Expected Monthly Additional Revenue	₹5,000

Net Monthly Cash Flow
Formula: `=K14-K13`

Financial Decision Model – Additional Counter	
Initial Counter Investment	50000
Monthly Operating Cost	8000
Expected Monthly Additional R	12395.79436

Figure 8.8: Financial Decision Model – Additional Counter

8.10 NPV Analysis

The **NPV (Net Present Value)** function was used to evaluate the projected cash flows of the additional counter using a 10% annual discount rate.

NPV – Additional Counter
Formula: `=NPV(10%/12,(K14-K13)*12)-K12`
Result: **₹-85,702.48**

NPV – Additional Counter	₹ 2,313.59

Figure 8.9: NPV Analysis Result

8.11 IRR Analysis

The **IRR (Internal Rate of Return)** function was applied to the projected cash flows.
The initial cash-flow structure did not provide a positive cash flow required for calculating a conventional IRR.

IRR – Additional Counter
Formula: `=IFERROR(IRR(CHOOSE({1,2,3,4,5,6,7,8,9,10,11,12,13},-K12,(K14-K13),(K14-K13),(K14-K13),(K14-K13),(K14-K13),(K14-`

K13),(K14-K13),(K14-K13),(K14-K13),(K14-K13),(K14-K13),(K14-K13))),"N/A - No Positive Cash Flow")
Result: 1%

IRR – Additional Counter	1%

Figure 8.10: IRR Analysis Result

8.12 PMT – Financing Analysis

The **PMT** function was used to calculate the monthly financing payment for the additional counter.

Monthly Financing Payment
Formula: *=PMT(10%/12,12,-K12)*
Result: **₹4,395.79**

Annual Financing Cost
Formula: *=K18*12*
Result: **₹52,749.53**

Analysis	Result
Monthly Financing Payment	₹ 4,395.79
Annual Financing Cost	52749.53234

Figure 8.11: PMT Financing Analysis

8.13 Goal Seek – Break-Even Analysis

Excel Goal Seek was used to determine the monthly revenue required to achieve a net monthly cash flow of zero.

The Goal Seek configuration was:

Parameter	Value
Set Cell	Net Monthly Cash Flow
Target Value	₹0
Changing Cell	Expected Monthly Additional Revenue

Required Monthly Revenue
Formula: =K13+K18
Result: **₹12,395.79**

Net Monthly Cash Flow
Formula: =K14-K13-K18
Result after Goal Seek: **₹0.00**

Revenue Break-Even Analysis	
Required Monthly Revenue	₹ 12,395.79
Net Monthly Cash Flow	₹ 0.00

Figure 8.12: Goal Seek Break-Even Analysis

8.14 Module 2 Outcome

Module 2 transformed the cleaned QueueFlow dataset into analytical, forecasting and decision-support information. Daily and monthly queue patterns were analysed using queue count and average waiting time.

Forecasting functions including **FORECAST, TREND, RSQ and LINEST** were used to estimate future waiting time and analyse trends. Date functions including **WORKDAY, NETWORKDAYS and DATEDIF** were used for operational period analysis.

Financial functions such as **NPV, IRR and PMT** were applied to an additional counter model, while **Goal Seek** was used to determine the revenue required to achieve break-even.

The results generated in Module 2 will be used as supporting inputs for **Module 3 – Power Query & What-If Analysis**.

MODULE 3 – DATA VALIDATION, ERROR HANDLING, POWER QUERY & WHAT-IF ANALYSIS

9.1 Data Validation

Data Validation was used to control the values entered in the **Priority** field. A list-based validation was applied so that only predefined priority categories could be selected.

The validation list contains **Low, Normal, High and Urgent**. This helps maintain consistency in the queue dataset and prevents invalid priority values.

Validation List: `Low, Normal, High, Urgent`
Result: Priority values are restricted to the four predefined categories.

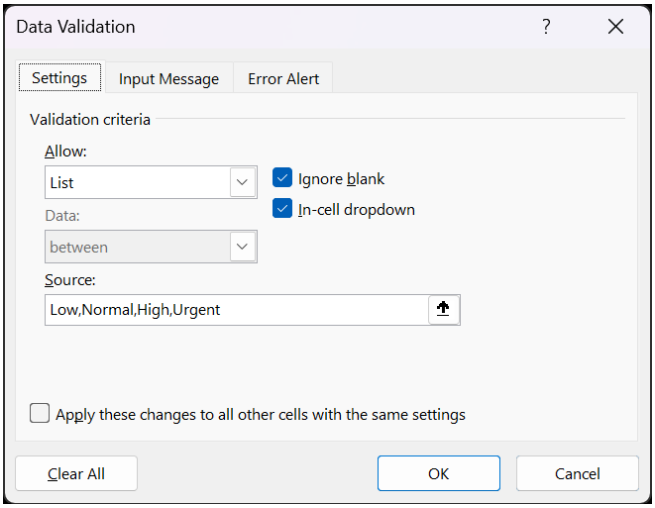


Figure 9.1: Data Validation applied to Priority column

9.2 Conditional Formatting

Conditional Formatting was applied to identify abnormal and invalid values in the queue dataset. Waiting Time, Service Time and Satisfaction Score were checked using conditional formatting rules.

The following conditions were highlighted:

- Waiting Time greater than 20 minutes
- Negative Waiting Time
- Service Time greater than 20 minutes
- Negative Service Time
- Satisfaction Score below 1 or above 5

This makes unusual records easier to identify during data validation.
Formula: `=L2>20`
Formula: `=L2<0`
Formula: `=M2>20`
Formula: `=M2<0`
Formula: `=OR(R2<1,R2>5)`

Result: Invalid and abnormal queue records were highlighted automatically.

Location_ID	Priority	Customer_Age	Waiting_Time	Service_Time	Status	Payment_Mode	Service_Fee	Staff_ID	Satisfaction_Score
LOC02	Normal	28	15	15	Completed	Card	250	STF0203	5
LOC01	Urgent	68	12	7	Completed	Card	100	STF0106	4
LOC05	High	54	3	5	Cancelled	Cash	0	STF0502	5
LOC01	Normal	64	12	7	Completed	Card	0	STF0101	5
LOC01	Normal	46	0	22	Completed	Cash	250	STF0101	5
LOC02	Low	68	1	19	Completed	UPI	250	STF0203	3
LOC05	High	26	16	5	Completed	N/A	100	STF0504	5
LOC03	Normal	32	6	5	Completed	Card	100	STF0304	4
LOC06	Normal	41	0	9	Completed	N/A	100	STF0603	4
LOC06	Normal	73		9	Completed	N/A	150	STF0601	4
LOC06	Normal	66	6	7	No Show	Cash	0	STF0604	5
LOC04	Urgent	18	11	7	Completed	UPI	100	STF0402	1
LOC01	Normal	44	6	9	Completed	UPI	200	STF0106	3
LOC06	Normal	56	21	5	Pending	Card	100	STF0601	1
LOC06	High	18	10	6	No Show	Online	100	STF0601	2
LOC03	Normal	69	18	6	Completed	Online	100	STF0302	2
LOC01	Normal	40	17	6	Completed	Online	0	STF0102	2
LOC02	Normal	71	8	8	Completed	Online	100	STF0202	5
LOC04	Normal	38	8	9	Completed	N/A	150	STF0401	1
LOC04	Normal	60	0	6	Completed	UPI	100	STF0403	1
LOC06	Normal	45	6	9	Completed	N/A	150	STF0604	1
LOC04	High	32	8	3	Completed	Card	100	STF0405	5
LOC04	Normal	60	12	13	Completed	N/A	0	STF0405	3
LOC03	Normal	60	3	10	No Show	Card	150	STF0306	2
LOC03	Normal	53	0	18	Completed	Cash	200	STF0305	4
LOC02	High	23	8	5	Completed	Card	100	STF0204	1
LOC01	Normal	18	19	8	Completed	Online	100	STF0105	4
LOC03	Normal	48	14	9	Completed	Cash	0	STF0302	3
LOC01	Normal	19	9	9	Pending	Online	200	STF0104	1
LOC06	Normal	47	0	8	Completed	UPI	100	STF0604	3
LOC04	Low	25	13	11	Completed	Online	200	STF0405	3
LOC02	Normal	67	11	8	Completed	N/A	100	STF0201	3
LOC06	Normal	27	13	3	Completed	Cash	0	STF0601	2
LOC06	Normal	59	12	11	Completed	Online	0	STF0603	1
LOC04	Normal	38	12	5	Completed	Card	0	STF0405	2
LOC03	Urgent	71	6	4	Cancelled	UPI	200	STF0304	4
LOC01	Normal	19	19	23	Completed	n/a	250	STF0105	1
LOC04	Normal	66	9	8	Completed	Cash	250	STF0401	5
LOC02	NORMAL	42	2	12	Completed	Online	200	STF0201	5
loc01	Normal	35	16	13	Completed	Cash	0	STF0104	1
LOC03	Normal	44	11	7	Completed	Card	100	STF0304	5
LOC01	Normal	62	16	9	Completed	Online	200	STF0106	5
LOC05	High	57	5	8	Completed	Cash	150	STF0505	1

Figure 9.2: Conditional Formatting applied to queue records

9.3 Error Handling

Excel error-handling functions were used to manage lookup and calculation errors. XLOOKUP was tested with both valid and invalid Service IDs. IFERROR and IFNA were used to display meaningful messages instead of Excel errors.

XLOOKUP – Valid Service ID

Formula: `=XLOOKUP("S001",Service_Master!A:A,Service_Master!B:B,"Not Found")`

Result: Bill Payment

XLOOKUP – Invalid Service ID

Formula: `=XLOOKUP("S999",Service_Master!A:A,Service_Master!B:B,"Not Found")`

Result: Not Found

IFERROR – Calculation Error

Formula: `=IFERROR(100/0,"Error Handled")`

Result: Error Handled

IFNA – Lookup Error

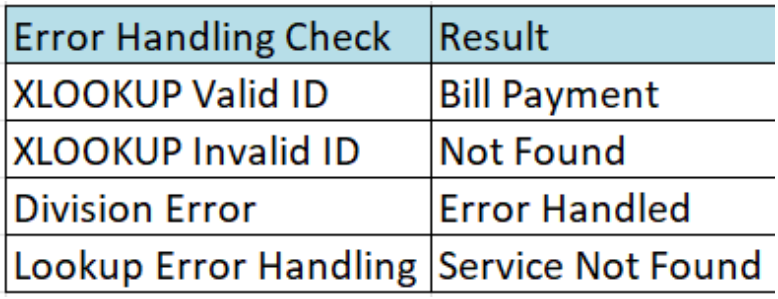
Formula:

`=IFNA(INDEX(Service_Master!B:B,MATCH("S999",Service_Master!A:A,0)),"Service Not Found")`

Result: Service Not Found

Error Handling Results

Error Handling Check	Result
XLOOKUP Valid ID	Bill Payment
XLOOKUP Invalid ID	Not Found
Division Error	Error Handled
Lookup Error Handling	Service Not Found



Error Handling Check	Result
XLOOKUP Valid ID	Bill Payment
XLOOKUP Invalid ID	Not Found
Division Error	Error Handled
Lookup Error Handling	Service Not Found

Figure 9.3: Error Handling Analysis Results

9.4 Formula Auditing

Formula Auditing was used to verify important calculations and trace the cells used by formulas. The main queue metrics were checked using Excel functions.

Total Queue Records

Formula: `=COUNTA(Cleaned_Data!A2:A10001)`
Result: 10,000

Average Waiting Time

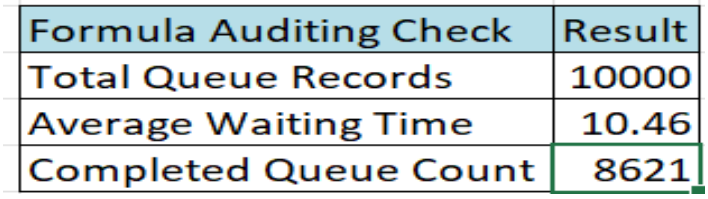
Formula: `=AVERAGE(Cleaned_Data!L2:L10001)`
Result: 10.46 minutes

Completed Queue Count

Formula: `=COUNTIF(Cleaned_Data!N2:N10001,"Completed")`
Result: 8,621

Formula Auditing Results

Metric	Result
Total Queue Records	10,000
Average Waiting Time	10.46 minutes
Completed Queue Count	8,621



Formula Auditing Check	Result
Total Queue Records	10000
Average Waiting Time	10.46
Completed Queue Count	8621

Figure 9.4: Formula Auditing and Analysis Results

9.5 Power Query – Data Cleaning

Power Query was used to import and transform the raw queue dataset. The original `Raw_Queue_Data` sheet was kept unchanged while the data was cleaned and transformed through Power Query.

The following transformations were performed:

- Imported data using **From Table/Range**
- Removed duplicate Queue IDs
- Standardized Customer Type values
- Standardized Status values
- Standardized Payment Mode values
- Converted Service ID, Counter ID and Location ID to uppercase
- Changed numerical columns to appropriate data types
- Removed errors from selected columns
- Replaced null values where required

After the transformation process, **9,957 records** were loaded into the `PowerQuery_Data` sheet.

Result: Cleaned and standardized data was successfully loaded using Power Query.

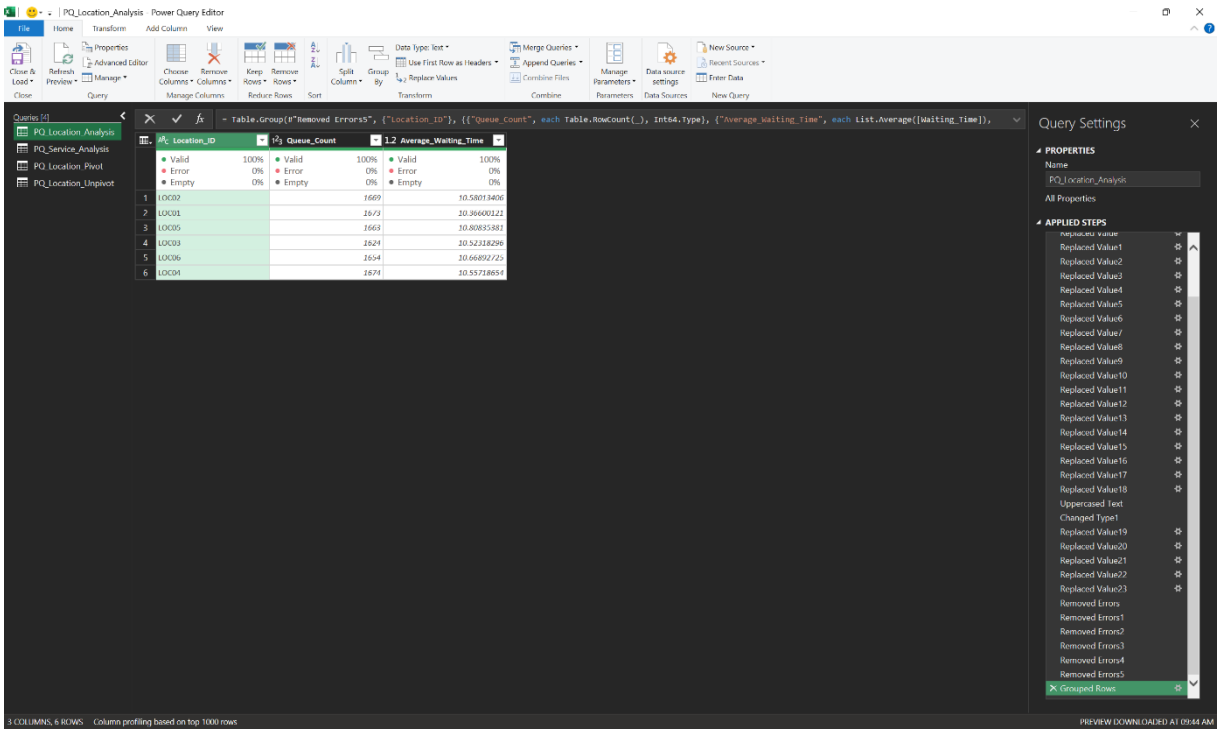


Figure 9.5: Power Query Data Cleaning and Transformation

9.6 Power Query – Group By, Sort and Filter

Power Query **Group By** was used to perform service-level analysis. The data was grouped according to `Service_ID`, and queue count and average waiting time were calculated.
The resulting service analysis was sorted according to queue count and filtered to identify services having an average waiting time greater than **10.5 minutes**.

Service Analysis

Queue Count

Formula/Calculation: `=Count of Service_ID`

Average Waiting Time

Formula/Calculation: `=Average of Waiting_Time`

Filter Condition: `=Average Waiting Time > 10.5 minutes`

Result: 7 service records satisfied the average waiting time filter.

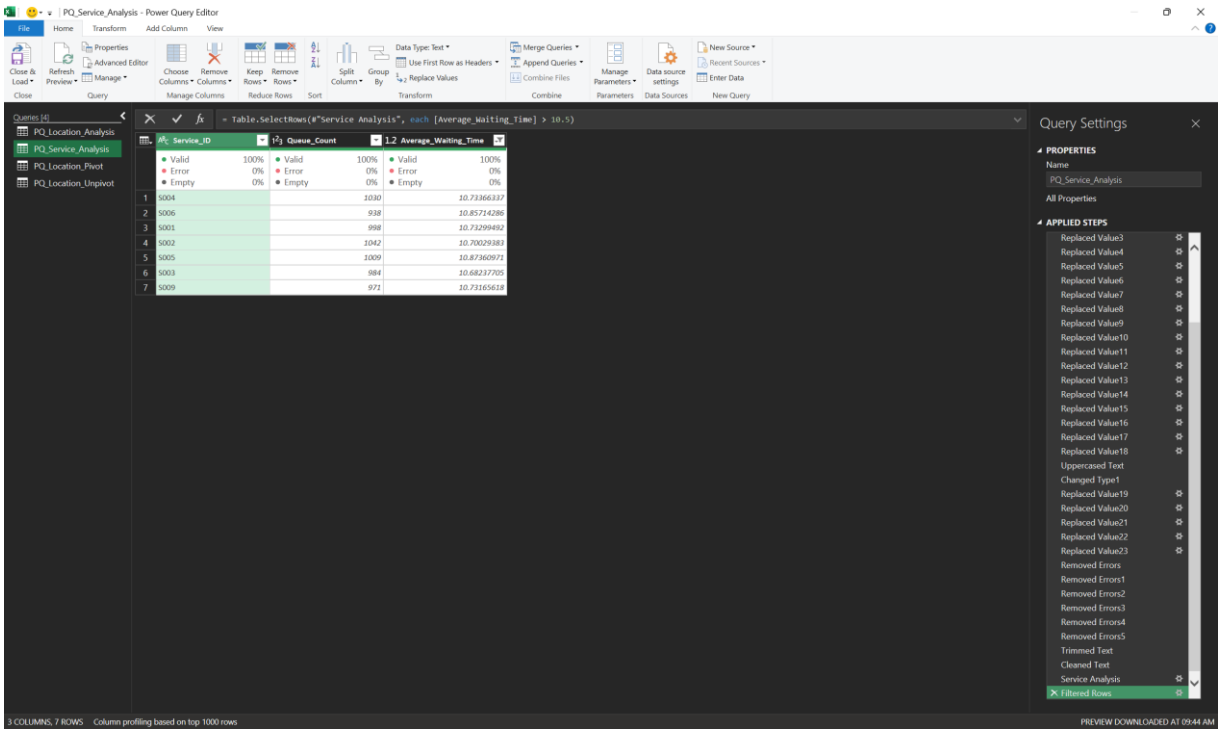


Figure 9.6: Power Query Service Grouping and Filtering

9.7 Power Query – Location Analysis

Power Query Group By was also used to analyse queue activity by location.
The data was grouped using `Location_ID`, and the number of queue records and average waiting time were calculated.

Location Analysis

Location ID	Location
LOC01	Central Branch
LOC02	Market Branch
LOC03	Station Branch
LOC04	City Branch
LOC05	Main Branch
LOC06	East Branch

Queue Count

Formula/Calculation: `=Count of Location_ID`

Average Waiting Time
Formula/Calculation: `=Average of Waiting_Time`
Result: Location-wise queue count and average waiting time were generated using Power Query.

Figure 9.7: Power Query Location Analysis

9.8 Power Query – Pivot and Unpivot

Power Query Pivot was used to convert the Location ID values into separate columns based on queue counts.
The resulting pivot table contained separate columns for:
`LOC01, LOC02, LOC03, LOC04, LOC05, LOC06`
The pivoted data was then Unpivoted to convert the location columns back into an **Attribute-Value** structure.

Pivot Result

Location	Queue Count
LOC01	1,673
LOC02	1,669
LOC03	1,624
LOC04	1,674
LOC05	1,663
LOC06	1,654

Result: Data was successfully transformed between wide and long formats using Pivot and Unpivot.

Figure 9.8: Power Query Pivot Transformation

Figure 9.9: Power Query Unpivot Transformation

9.9 Scenario Manager

Scenario Manager was used to analyse different expected monthly revenue conditions. Three scenarios were created using the same changing cell for comparison.

Scenario Analysis

Scenario	Expected Monthly Revenue
Low Revenue	₹8,000
Normal Revenue	₹12,000
High Revenue	₹16,000

The scenarios were created by changing the **Expected Monthly Revenue** value in cell B2.
Formula: `=B2-B8`
Where:

- B2 = Expected Monthly Revenue
- B8 = Monthly Operating Cost

Result: Three revenue scenarios were successfully created and compared using Scenario Manager.

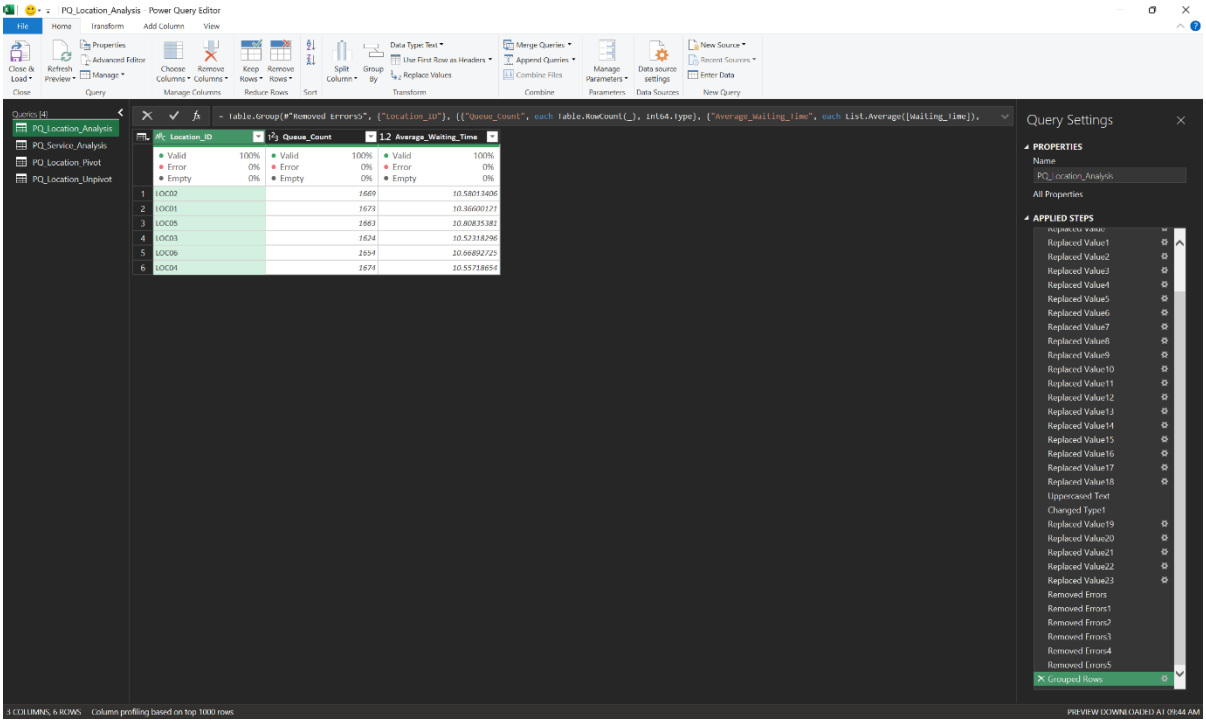


Figure 9.10: Scenario Manager Summary

9.10 Goal Seek – Break-Even Analysis

Goal Seek was used to determine the expected monthly revenue required to achieve a net monthly cash flow of zero.
The model used the following parameters:

Parameter	Value
Initial Investment	₹50,000
Monthly Operating Cost	₹8,000
Target Net Monthly Cash Flow	₹0

The Goal Seek configuration was:

Goal Seek Parameter	Setting
Set Cell	Net Monthly Cash Flow
Target Value	0
Changing Cell	Expected Monthly Revenue

Net Monthly Cash Flow

Formula: `=B2-B8`
Result: ₹0.00

Required Monthly Revenue

Result: ₹8,000
Therefore, the model reaches a zero net monthly cash flow when the expected monthly revenue is **₹8,000**.

Initial Investment	50000
Monthly Operating Cost	8000
Net Monthly Cash Flow	0
Expected Monthly Reven	8000

Figure 9.11: Goal Seek Break-Even Analysis

9.11 Data Table – Revenue Sensitivity Analysis

A one-variable Data Table was created to analyse the effect of different expected monthly revenue values on net monthly cash flow.

The operating cost was kept constant at **₹8,000** while the expected revenue was varied.

Net Monthly Cash Flow

Formula: `=B9`

Data Table Results

Expected Monthly Revenue	Net Cash Flow
₹6,000	-₹2,000
₹7,000	-₹1,000
₹8,000	₹0
₹9,000	₹1,000
₹10,000	₹2,000
₹12,000	₹4,000

The Data Table shows how changes in expected monthly revenue affect the resulting net monthly cash flow.

Result: The sensitivity analysis shows the relationship between expected revenue and net monthly cash flow.

Expected Monthly Revenue	0
6000	-2000
7000	-1000
8000	0
9000	1000
10000	2000
12000	4000

Figure 9.12: Data Table Revenue Sensitivity Analysis

9.12 Module 3 Conclusion

Module 3 focused on data validation, error handling, formula auditing, Power Query and What-If Analysis. Data Validation and Conditional Formatting helped maintain data quality and identify abnormal records. Error-handling functions such as XLOOKUP, IFERROR and IFNA were used to manage invalid lookup and calculation results.

Power Query was used to clean, transform, group, filter, pivot and unpivot the queue dataset. Scenario Manager, Goal Seek and Data Table were then used to perform What-If Analysis and examine the relationship between expected revenue and net monthly cash flow.

MODULE 4 – POWER PIVOT, DAX & DASHBOARD

10.1 Module Overview

Module 4 of the QueueFlow Analytics project focuses on **Power Pivot, DAX, data modelling, KPI creation and dashboard development**. The cleaned queue dataset was added to the Excel Data Model and connected with the Service Master table using a relationship based on `Service_ID`.

DAX measures and calculated columns were then created to analyse queue performance, waiting time, service revenue, customer satisfaction and high-wait cases.

10.2 Power Pivot Data Model

The `Cleaned_Data_Model` table containing the queue transaction data was added to the Excel Data Model. The `Service_Master` table was also added to provide service-level information.

A relationship was established between the two tables using `Service_ID`.

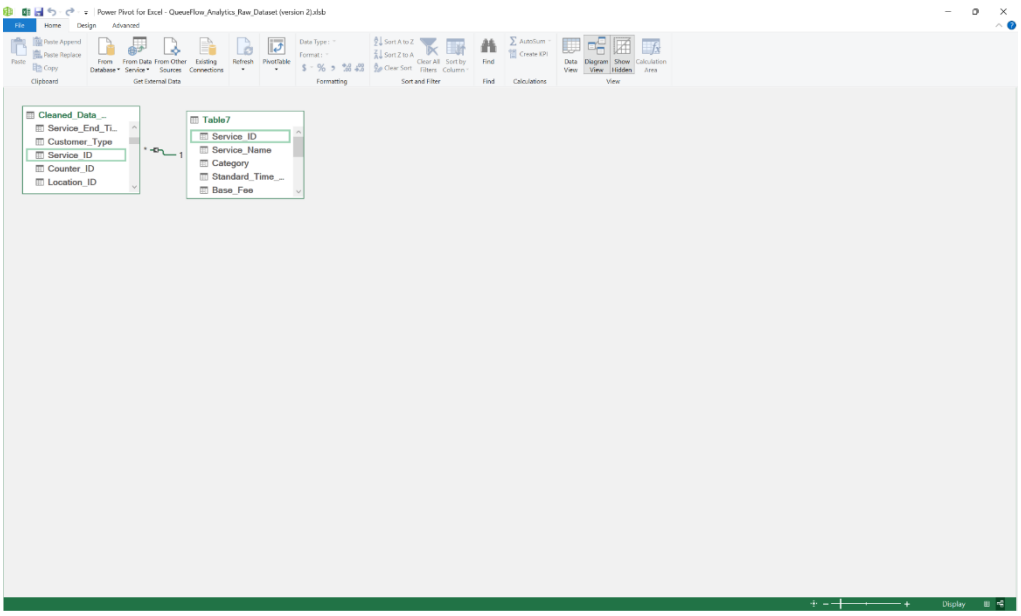


Figure 10.1: Power Pivot Data Model

10.3 Relationship Between Tables

The relationship between the queue transaction table and the service master table allows service-related information to be connected with individual queue records.

The relationship was created as:
Service_Master → Cleaned_Data_Model
`Service_ID` was used as the common field.

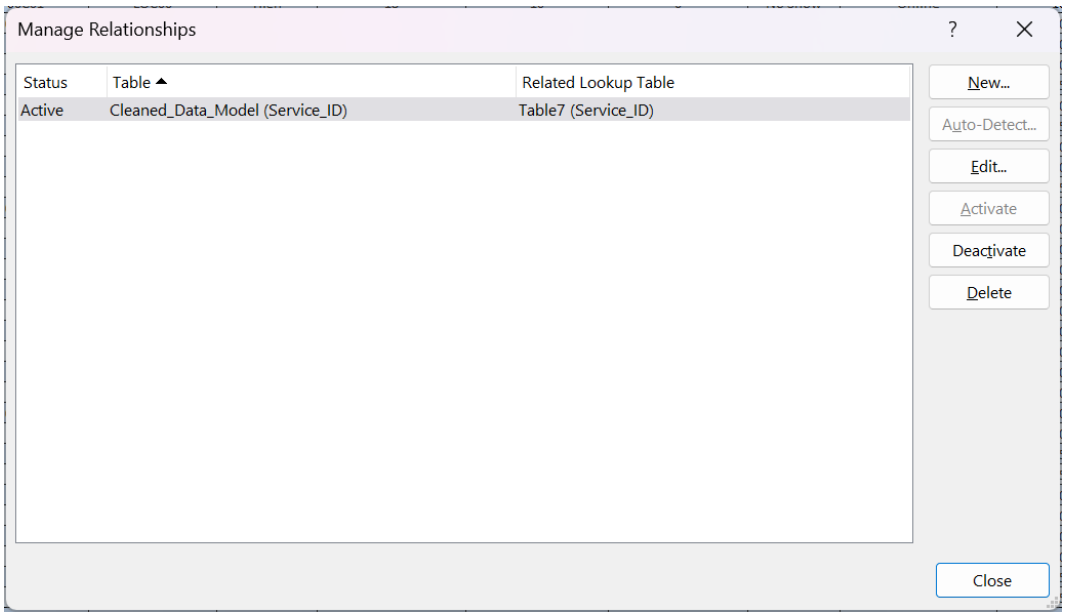


Figure 10.2: Relationship Between Service Master and Queue Data

10.4 DAX Measures

Several DAX measures were created to calculate important QueueFlow performance metrics. The measures include:

- Total Queue Count
- Completed Queue Count
- High-Wait Queue Count
- Average Waiting Time
- Average Service Time
- Average Satisfaction Score
- Total Service Revenue
- Average Service Fee
- Completed Revenue
- Total Queue Load
- High Wait Revenue

These measures provide dynamic calculations that can be used in PivotTables and the final dashboard.

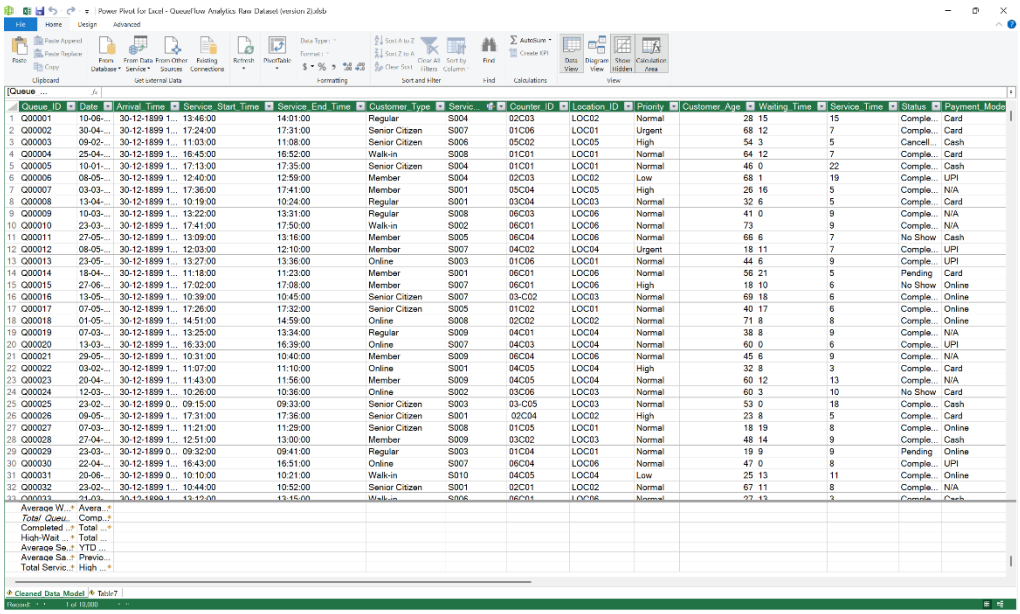


Figure 10.3: DAX Measures Created in Power Pivot

10.5 Average Waiting Time using DAX

The waiting-time field required conversion before performing the average calculation in the Data Model. Therefore, `AVERAGEX` and `VALUE` were used to calculate the average waiting time.

The resulting overall average waiting time was approximately **10.46 minutes**.

Average Waiting Time

Formula:

```
=AVERAGEX(Cleaned_Data_Model,IFERROR(VALUE(Cleaned_Data_Model[Waiting_Time]
),BLANK()))
```

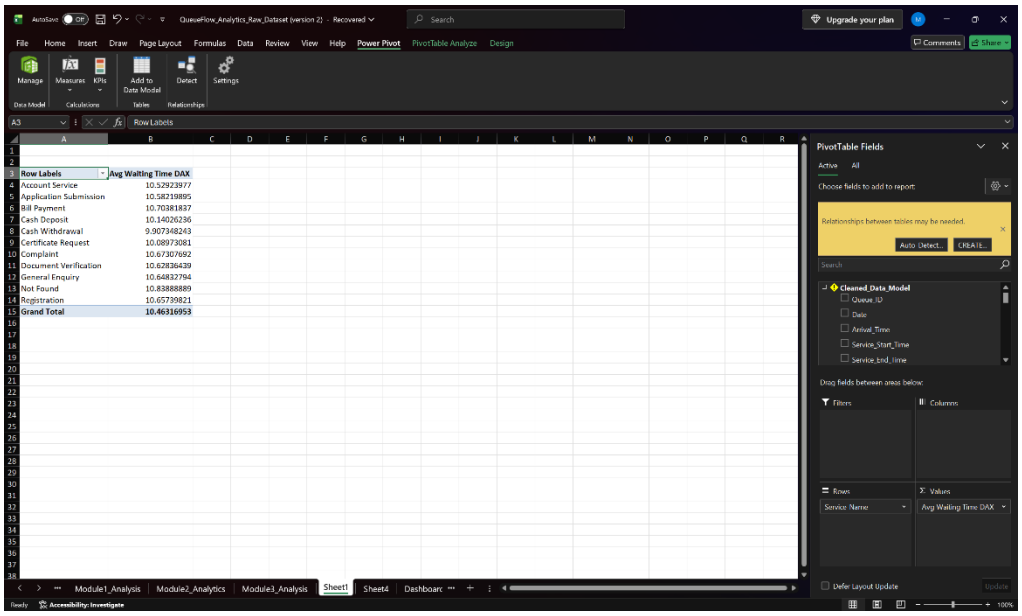


Figure 10.4: Service-wise Average Waiting Time using DAX

10.6 Queue Count Analysis

The `COUNTROWS` and `CALCULATE` functions were used to create queue-count measures.

Total Queue Count
Formula: `=COUNTROWS(Table6)`

Completed Queue Count
Formula:
```
=CALCULATE(COUNTROWS(Table6),Table6[Status]="Completed")
```

These measures provide dynamic information about the total and completed queue transactions.

10.7 High-Wait Queue Analysis

A DAX measure was created to identify queue records where the waiting time exceeded 20 minutes.

High-Wait Queue Count
Formula:
```
=CALCULATE(COUNTROWS(Table6),Table6[Waiting_Time]>20)
```

This metric helps identify higher waiting-time cases for operational monitoring.

HIGH WAIT QUEUE COUNT	739

Figure 10.5: High-Wait Queue Analysis

10.8 Revenue Analysis using DAX

DAX measures were created to calculate total service revenue, average service fee and completed revenue.

Total Service Revenue
Formula: `=SUM(Table6[Service_Fee])`

Average Service Fee
Formula: `=AVERAGE(Table6[Service_Fee])`

Completed Revenue
Formula: `=CALCULATE(SUM(Table6[Service_Fee]),Table6[Status]="Completed")`

These measures provide dynamic revenue indicators for further analysis.

10.9 SUMX Analysis

The `SUMX` function was used for row-level calculations before aggregation.
Total queue load was calculated by combining waiting time and service time.

Total Queue Load
Formula: `=SUMX(Table6,Table6[Waiting_Time]+Table6[Service_Time])`

A SUMX-based service-fee calculation was also created.

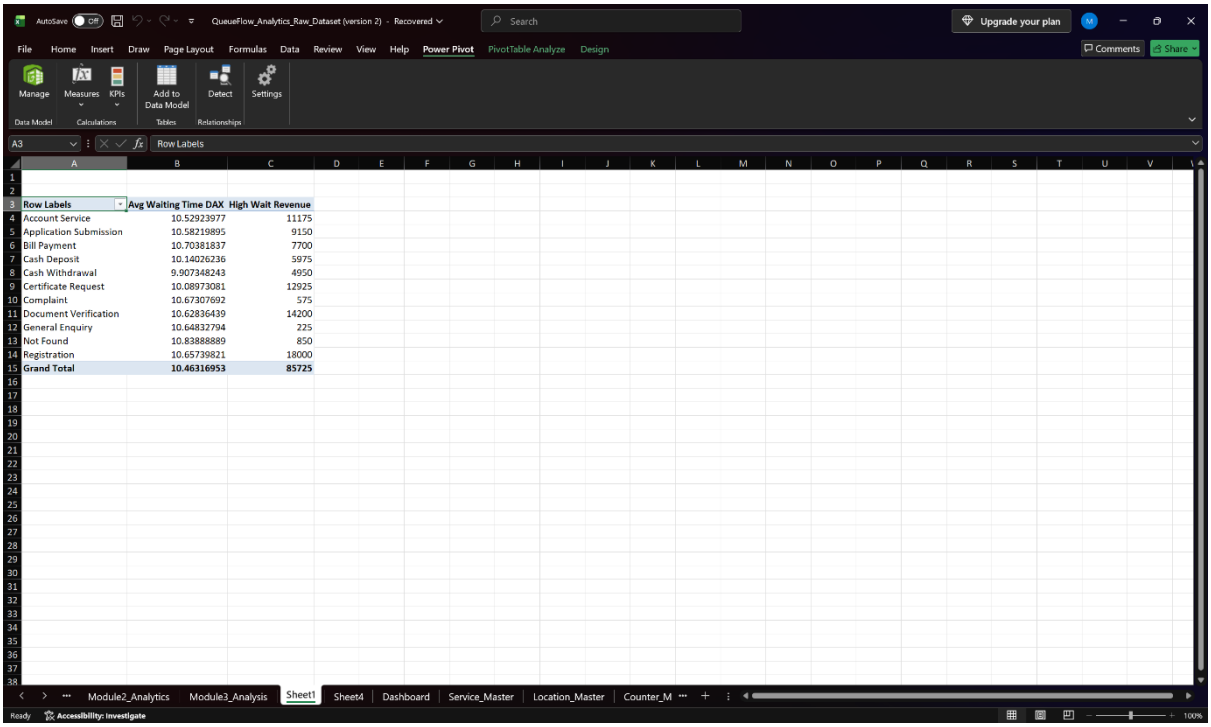
Total Service Fee SUMX
Formula: `=SUMX(Cleaned_Data_Model,Cleaned_Data_Model[Service_Fee])`

10.10 FILTER and High-Wait Revenue

The `FILTER` function was combined with `CALCULATE` to calculate revenue generated by transactions having waiting time greater than 20 minutes.

High Wait Revenue
Formula:

```
=CALCULATE(SUM(Cleaned_Data_Model[Service_Fee]),FILTER(Cleaned_Data_Model,C
leaned_Data_Model[Waiting_Time]>20))
```



Row Labels	Avg Waiting Time DAX	High Wait Revenue
Account Service	10.52923977	11175
Application Submission	10.58219895	9150
Bill Payment	10.70381837	7700
Cash Deposit	10.14026236	5975
Cash Withdrawal	9.907348243	4950
Certificate Request	10.08973081	12925
Complaint	10.67307692	575
Document Verification	10.62836439	14200
General Enquiry	10.64832794	225
Not Found	10.83888889	850
Registration	10.65739821	18000
Grand Total	**10.46316953**	**85725**

Figure 10.6: High-Wait Revenue Analysis

10.11 RELATED Function

The `RELATED` function was used to retrieve the service category from the Service Master table. A calculated column named **Service Category** was created in the queue data model.

Service Category
Formula:

```
=RELATED(Table7[Category])
```

This allows queue records to be analysed according to service category.

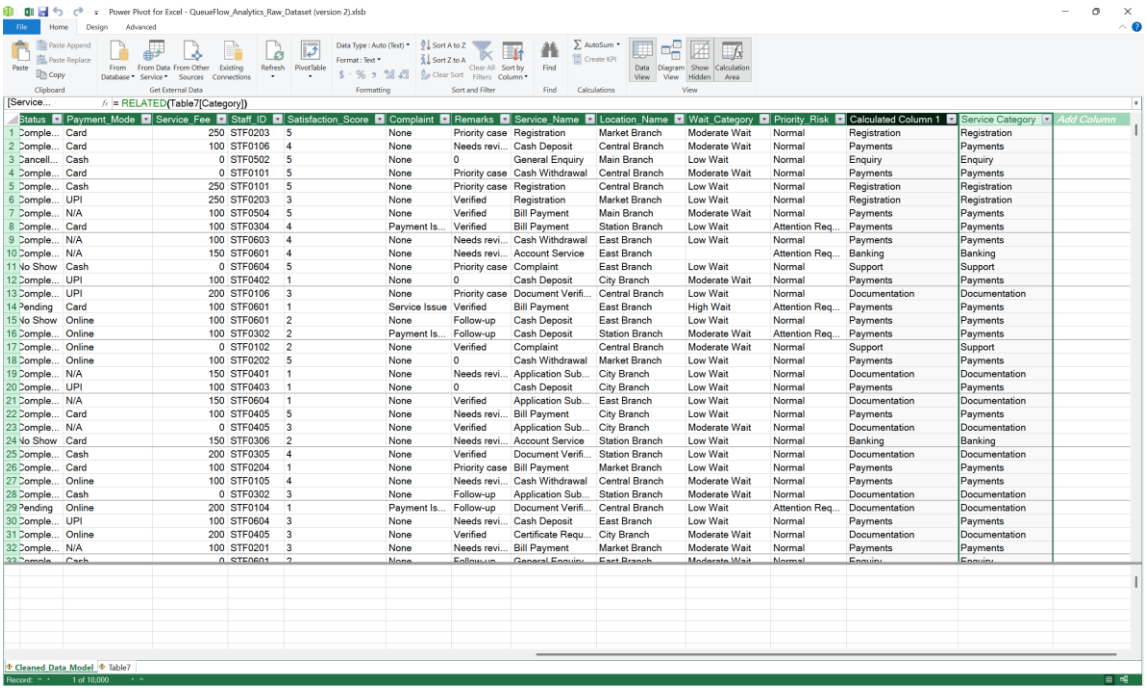


Figure 10.7: Service Category Calculated Column

10.12 Time Intelligence

Time-intelligence DAX functions were used to create revenue-based time calculations. The `TOTALYTD` function was used to calculate year-to-date revenue.

YTD Revenue

Formula: `=TOTALYTD([Completed Revenue DAX],Table7[Date])`

The `SAMEPERIODLASTYEAR` function was also used for previous-period comparison.

Previous Year Revenue

Formula: `=CALCULATE([Completed Revenue DAX],SAMEPERIODLASTYEAR(Table7[Date]))`

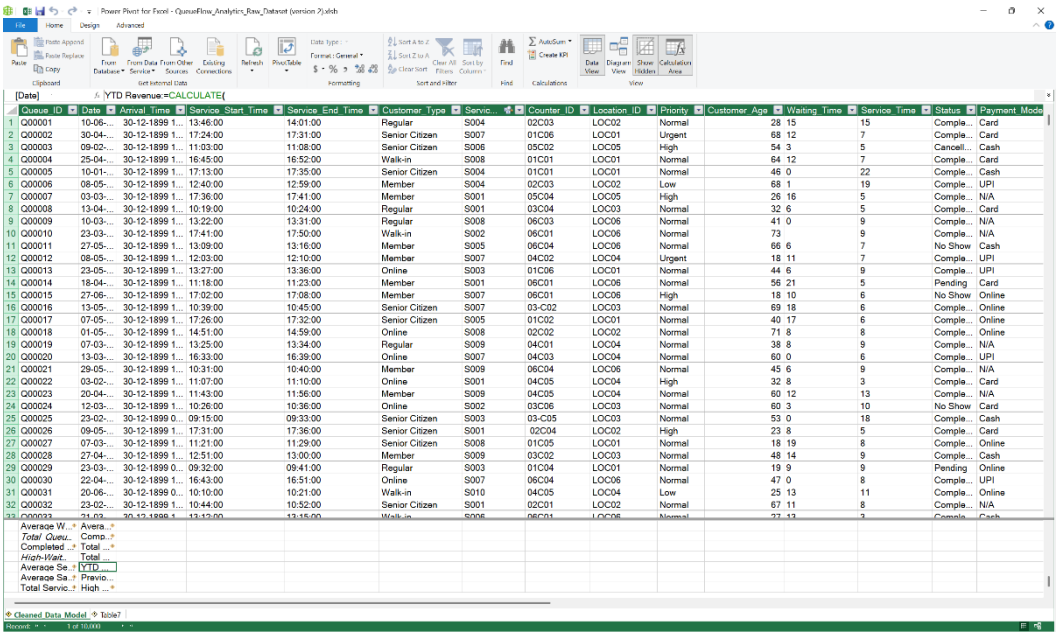


Figure 10.8: DAX Time-Intelligence Measures

10.13 KPI Creation

A KPI was created using the High Wait Percentage measure to monitor the proportion of queue records with waiting time greater than 20 minutes.

High Wait %

Formula:

```
=DIVIDE(CALCULATE(COUNTROWS(Cleaned_Data_Model),Cleaned_Data_Model[Waiting_
Time]>20),COUNTROWS(Cleaned_Data_Model))
```

The KPI was configured with a **10% target**.

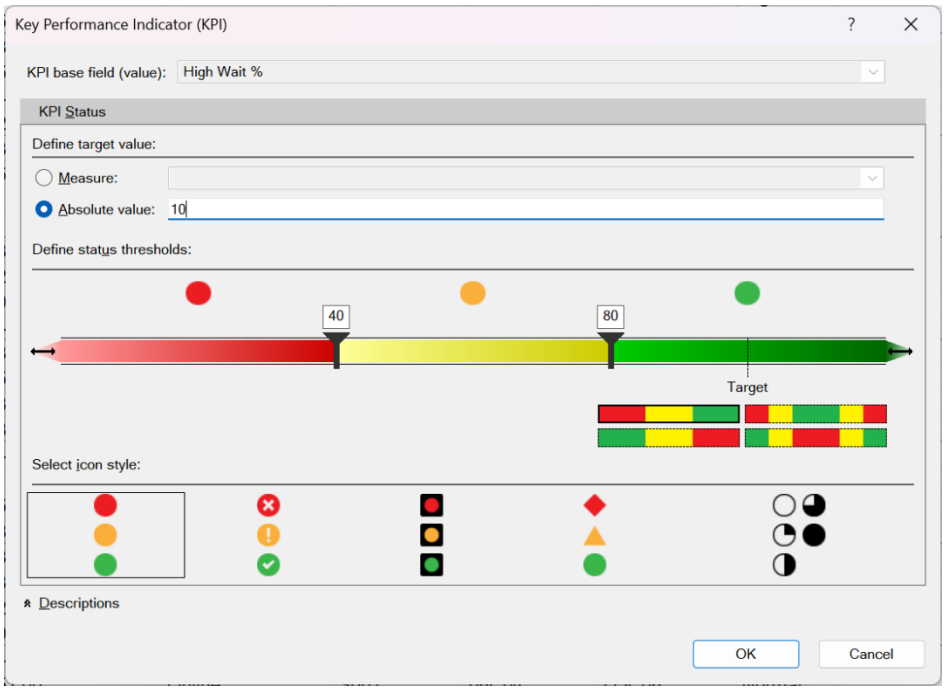


Figure 10.9: High-Wait Percentage KPI

10.14 Dashboard Development

The QueueFlow dashboard is designed to provide a consolidated view of the major queue-performance indicators.
The dashboard will include KPI-based metrics such as:

- Total Queue Records
- Average Waiting Time
- Completed Queue Count
- High-Wait Queue Count
- Average Satisfaction
- Total Service Revenue

The dashboard will also use PivotTables and visualizations to present queue and service-level performance.

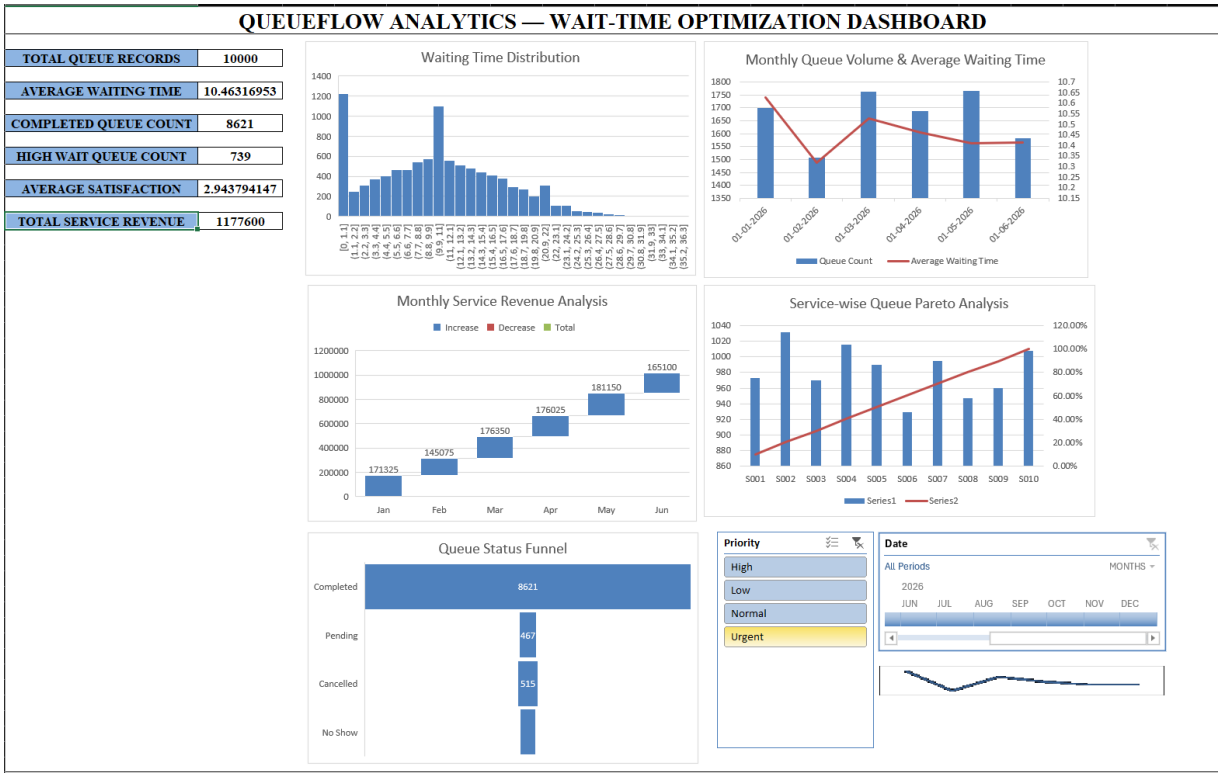


Figure 10.10: QueueFlow Analytics Dashboard

10.15 Interactive Dashboard Features

Interactive filtering can be added using **Slicers and Timeline controls** to allow the user to analyse queue performance based on service, location, priority and date.

These controls make the dashboard easier to explore and support interactive analysis of the QueueFlow dataset.

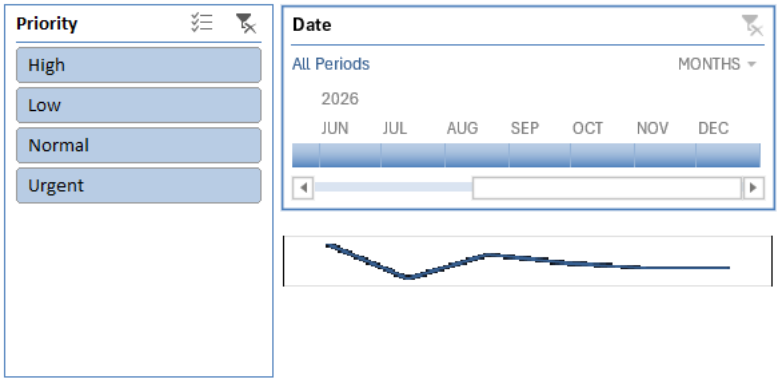


Figure 10.11: Dashboard Slicers and Timeline

10.16 Module 4 Outcome

Module 4 transformed the QueueFlow dataset into a relational analytical model using **Power Pivot and DAX**. Relationships were created between the queue data and Service Master, while DAX measures and calculated columns were used for queue, waiting-time, revenue and service-category analysis.

The KPI and dashboard components provide a consolidated view of QueueFlow performance and complete the analytical workflow developed throughout the project.

RESULTS & FINDINGS

The QueueFlow Analytics project successfully transformed raw and inconsistent queue data into a structured analytical dataset using Advanced Excel. The project covered data cleaning, validation, lookup operations, dynamic arrays, logical functions, forecasting, financial modelling, Power Query, Power Pivot, DAX and interactive dashboard development.

The analysis of **10,000 queue records** produced the following key findings:

- The overall **average waiting time was approximately 10.46 minutes**.
- The maximum recorded waiting time was **36 minutes**.
- **8,621 queue records** had a Completed status.
- **719 records** had a waiting time greater than 20 minutes and were classified as high-wait cases.
- The overall **average customer satisfaction score was approximately 2.94**.
- Total service revenue generated from the dataset was **₹11,77,600**.
- High-wait queue records were associated with approximately **₹85,725** in service revenue.
- Monthly analysis showed variations in queue volume and average waiting time across January to June 2026.
- Forecasting functions such as **FORECAST, TREND, RSQ and LINEST** were used to study the waiting-time trend.
- What-If Analysis using **Scenario Manager, Goal Seek and Data Tables** demonstrated how changes in revenue and operating conditions affect financial outcomes.
- Power Pivot and DAX enabled service-wise waiting-time analysis, high-wait analysis, revenue analysis and KPI creation.
- The final dashboard combined KPIs, charts, slicers, timeline controls and sparklines into a single interactive analytical view.

Overall, the findings demonstrate that Advanced Excel can be used not only for basic calculations but also for complete data preparation, analytical modelling, visualization and operational decision support.

CONCLUSION

The **QueueFlow Analytics – Wait-Time Optimization** project successfully demonstrated the application of Advanced Excel for analysing and optimizing queue waiting times. Starting with raw and inconsistent queue data, the project progressively applied data cleaning, validation, lookup functions, dynamic arrays and logical functions to prepare reliable analytical data.

Advanced analytical techniques such as forecasting, financial modelling, Power Query, Power Pivot and DAX were then applied to extract meaningful insights from the dataset. What-If Analysis techniques such as Scenario Manager, Goal Seek and Data Tables were also used to evaluate different operational and financial situations.

The final interactive dashboard provided a consolidated view of queue volume, waiting time, service revenue, customer satisfaction and high-wait cases. The use of slicers, timelines, KPIs and multiple visualization techniques made the analysis easier to understand and interact with.

Thus, the project demonstrates how **Advanced Excel can support the complete data analytics workflow — from raw data cleaning and transformation to analysis, visualization and decision-making.**